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GEOGRAPHICAL INFORMATION SYSTEMS FOR EARLY PREDICTION OF TTCL'S FIBER-OPTIC NETWORK FAULTS

A Case Study of Dodoma Municipal

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BSc Geographical Information Systems and Remote Sensing

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A Case Study of Dodoma Municipal

By

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A Dissertation submitted to the Department of Geospatial Sciences and Technology in partial fulfillment of the requirements of the award of Bachelor of Science in Geographical Information Systems and Remote Sensing (BSc. GIS & RS) of Ardhi University.

CERTIFICATION

The undersigned certify that they have proof read and hereby recommend for acceptance by the Ardhi University dissertation titled **"Geographical Information Systems for Early Prediction of TTCL's Fiber-Optic Network Faults: A case study of Dodoma Municipal."** in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Geographical Information Systems and Remote Sensing at Ardhi University.

.....

Dr. Joseph Hayola.

(Supervisor)

Date .....

DECLARATION AND COPYRIGHT

I, Shemng'ombe, Augustino Abraham, hereby declare that the contents in this dissertation are the results of my own findings through my study and investigation, and to the best of my knowledge. It has not been presented to anywhere else as a dissertation for diploma, degree or any similar academic award in any institution of higher learning.

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SHEMNG'OMBE, AUGUSTINO ABRAHAM

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DEDICATION

I dedicate this dissertation to my lovely family. My dear parents; Abraham Augustino and Rahel Abraham, have been a constant support and encouragement during the good and the challenging days in school and life in general. Through your prayers, support, and encouraging words I can stand as a proud knowledgeable person. I also dedicate this dissertation to my friends and course mates, who have supported and encouraged me to do better daily.

ABSTRACT

This research explores the application of Geographic Information Systems in predicting network faults whereabouts within the Tanzania Telecommunication Corporation Limited infrastructure in Dodoma Municipal. The current reactive fault management approach at TTCL, which basically relies on customer complaints and manual inspections, has led to service delays and dissatisfaction. This study implements a proactive GIS-based fault prediction approach by applying a Generalized Linear Model (GLM) that integrates historical fault records with spatial and environmental data such as elevation, rainfall, soil moisture, and temperature. Spatial analysis and logistic regression modelling were conducted using R Studio, QGIS, as well as Arc GIS software to identify areas at high risk of faults in both aerial and underground fiber-optic cables. The model implemented achieved a high predictive performance (AUC = 0.979), validating its effectiveness. Key risk factors identified include intense rainfall and extreme temperatures. This GIS-driven approach offers TTCL a valuable decision-support tool to enhance network reliability, improve maintenance efficiency, and reduce service disruptions which therefore contributes significantly to proactive infrastructure management in Tanzania's telecommunications sector.

TABLE OF CONTENTS

CERTIFICATION i
DECLARATION AND COPYRIGHT ii
ACKNOWLEDGEMENT iii
DEDICATION iv
ABSTRACT v
TABLE OF CONTENTS vi
LIST OF FIGURES ix
LIST OF TABLES x
LIST OF ACRONYMS AND ABBREVIATIONS xi
CHAPTER ONE 1
INTRODUCTION 1
1.1 Overview 1
1.2 Background of the study 1
1.3 Statement of the problem 2
1.4 Research Objectives 3
1.4.1 Main Objective 3
1.4.2 Specific Objectives 3
1.5 Research Questions 3
1.6 Significance of the study 3
1.7 Beneficiaries 4
1.8 Scope and Limitations 4
1.9 Dissertation Organization 5
CHAPTER TWO 6
LITERATURE REVIEW 6
2.1 Overview 6
2.2 Tanzania Telecommunications Corporation Limited 6
2.3 Fault Prediction Approaches 7
2.4 Modelling Approaches in Fault Prediction: GLM and ROC Analysis 7
2.5 Network Faults in Telecommunication Infrastructure 8
2.6 Geographical Information System 8
2.7 Spatial and Environmental Risk factors 9
2.8 Spatial Analysis 9
2.8.1 Spatial Analysis Techniques 10

2.9 Accuracy Assessment 11
CHAPTER THREE 13
METHODOLOGY 13
3.1 Overview 13
3.2 Description of the study area 13
3.3 Methodology Workflow 14
3.4 Data Collection 15
3.5 Data Pre-processing 16
3.6 Data Processing and analysis 21
3.7 Accuracy Assessment 23
3.7.1 ROC Curve and AUC Analysis 23
CHAPTER FOUR 25
RESULTS 25
4.1 Overview 25
4.2 Predictive Model Summary and Validation 25
4.2.1 Predictive Model Fit Statistics 27
4.3 Correlation Analysis 29
4.4 Terrain Mapping 31
4.5 Fault Risk Mapping 34
4.6 Buffer Analysis: Evaluating Proximity Zones of Interest. 35
4.7 Predicted Faults Locations 36
CHAPTER FIVE 37
DISCUSSION 37
5.1 Overview 37
5.2 Discussion on Terrain Mapping 37
5.3 Discussion on Predictive Model Performance and Accuracy 37
5.4 Discussion on Correlation Analysis 38
5.5 Discussion on the Fault Risk Map Visualization 38
5.6 Discussion on the Buffer Zones Around the Fiber-Optic Networks 39
5.7 Discussion on the Predicted Fault Locations 39
CHAPTER SIX 40
CONCLUSION AND RECOMMENDATION 40
6.1 Overview 40
6.2 Conclusion 40

6.3 Recommendation 40
REFERENCE 41
APPENDICES 44
APPENDIX 1 45
APPENDIX 2 48
APPENDIX 3 50
APPENDIX 4 52

LIST OF FIGURES

Figure 3.1: Dodoma Urban Location Map 14
Figure 3.2: Research Methodology Workflow 15
Figure 3.3: Visualization of Geocoded Fault Data in QGIS Software 18
Figure 3.4: Elevation Raster 18
Figure 3.5: Clipped Elevation Raster of Dodoma Urban 19
Figure 3.6: Network Infrastructure Layout 20
Figure 3.7: Selecting by Expression of Placement Types from the Attribute Table of Fiber 20
Figure 3.8: Pairwise Correlation & Distribution of Faults Vs Variables 23
Figure 3.9: ROC Curve for Predictive Model 24
Figure 3.10: ROC Curve for Predictive Model with AUC 24
Figure 4.1: Correlation Matrix of Fault Presence Vs Environmental Risk Factors 29
Figure 4.2: Scatterplots of Environmental Risk Factors Vs Fault Presence 30
Figure 4.3: Terrain Mapping With Slope Classification At Fault Locations 31
Figure 4.4: Box Plot of Slope Distribution in Fault Areas 32
Figure 4.5: A Histogram of Fault Occurrence by Terrain Slope 33
Figure 4.6: Fault Risk Map 34
Figure 4.7: Buffer Zones Around Aerial & Underground Fiber-Optic Cables 35
Figure 4.8: An Overlay of Real & Predicted Fault Locations 36

LIST OF TABLES

Table 3.1: Research Datasets 15
Table 3.2: Sample of Raw TTCL Fault Records 16
Table 3.3: TTCL Fault Records After Pre-processing 17
Table 4.1: Predictive Model Summary 26
Table 4.2: Predictive Model Fit Statistics 28

LIST OF ACRONYMS AND ABBREVIATIONS

AIC	Akaike's Information Criterion
AUC	Area Under the Curve
CRS	Coordinate Reference System
CSV	Comma -Separated Values
DEM	Digital Elevation Model
EPSG	European Petroleum Survey Group
GeoTIFF	Geographic Tagged Image File Format
GIS	Geographic Information System
GLM	Generalized Linear Model
IoT	Internet of Things
NA	Not Available /Missing Value
NetCDF	Network Common Data Form
OSM	Open Street Map
QGIS	Quantum Geographic Information System
ROC	Receiver Operating Characteristic
SRTM	Shuttle Radar Topography Mission
TT	Trouble Ticket Number
TTCL	Tanzania Telecommunication Corporation Limited
USGS	United States Geological Survey
UTM	Universal Transverse Mercator

CHAPTER ONE
INTRODUCTION

1.1 Overview

This chapter introduces the research context and the need for applying GIS in predicting network faults within the telecommunication infrastructure, specifically focusing on TTCL in Dodoma Municipal. It begins by outlining the background of the study, highlighting the challenges faced by TTCL in managing network faults through reactive approaches. Then it presents the problem statement, which emphasizes the need for a proactive approach through the implementation of predictive modelling. It further defines the research objectives, both main and specific, and formulates key research questions aiming at guiding the study. The significance of the research is also pointed out, emphasizing the benefits for TTCL, customers, and the broader telecommunications industry. In addition to this, the chapter identifies the primary beneficiaries of the study and shows its scope and limitations, particularly in terms of data access and geographical focus. Finally, it provides an outline of the dissertation structure to guide the reader through the subsequent chapters.

1.2 Background of the study

Telecommunications, as an information subsector, is an important sector in Tanzania and the world in general as it issues the provision of services, such as fixed-line and mobile internet, and the infrastructure that relates to the transmission of signals, messages and data. The significance of having a reliable telecommunication infrastructure is that it assists in both economic and social development in the sense that the country earns enough earnings from the sector to ensure other economic activities are well run and socially satisfy its people in accessing reliable information and services keeping up with technological changes that are daily taking place (United Republic of Tanzania, 2017). In Tanzania, one amongst the oldest and the most profound telecommunication company that is responsible for the provision of such services and infrastructure is the Tanzania Telecommunications Corporation Limited. TTCL ideal responsibility lies in ensuring that there’s a connection or a bond among people and their businesses through out the country. Having realising that there's a growing demand for reliable communication services, the company has been encountering a significant number of challenges in managing issues of network faults as their approach towards fault resolution lies in customer complaints and manual inspections by their technical teams (Semboko, 2015).

With such a reactive approach, being adopted over a number of years, issues the need for a better and more proactive approach which will ensure the reduction in service disruption

leading to fault existence and better satisfaction to the customers through proper maintenance of the infrastructures responsible for transmission of data and signals. Realising that the presence of network faults is highly attributed by TTCL's network of aerial and underground fiber-optic cables being vulnerable to the environmental and spatial risk factors. The aerial cables get damaged by wind, poles falling over, physical damage from falling trees, and even vandalism while underground cables suffer from issues like excavation damages, water getting into the ground, and aging of the infrastructures. Sometimes, issues like electricity power surges, electromagnetic interference, transmission problems affecting signal strength and data flow, as well as software or configuration issues obstructed from system bugs or cyber threats; they all contribute to the existence of network faults (Xu & Yuan, 2023).

The environmental risk factors such as unfavourable climatic conditions, and spatial risk factors like topography and terrain variations act as influential contributors to the presence of both aerial and underground network faults. These addressed factors are rarely being analysed in the required depth leaving exposed gaps in the company's fault management strategies. Therefore, the utilization of GIS technology will offer a practical solution through the use of tools in analysing spatial and environmental data making it possible to detect patterns and identify areas which are mostly at risk for faults before they occur making it easy to estimate the likelihood of fault occasions (Ghorbanzadeh et al., 2019).

With the identified potentials of GIS, still the use of it in the telecommunications fault management is not yet being fully explored, especially in Tanzania. Given the current researches, most of them as well as their applications, tend to focus on sectors like energy and utilities which leaves a gap in the telecommunications sector. Thus, this research will fill that gap through predictive modelling that is implemented specifically to address the challenge of network faults for TTCL as the model allows the integration of fault records, spatial and environmental data so as to identify the high-risk areas or zones within the network infrastructure and point out possible locations where network fault can occur, giving the company the advantage of reducing delays while improving its overall service reliability as well as assurance of retaining and improving customer satisfaction.

1.3 Statement of the problem

TTCL's reliance on reactive fault management methods has led to delayed responses, increased operational costs, and frequent service disruptions. Aerial and underground fiber-optic cable networks are exposed to environmental hazards and human caused damages and yet, the spatial

and environmental risk factors are not systematically analysed. The absence of a geospatial predictive system prevents TTCL from identifying areas that are highly susceptible to be affected by faults and therefore, resulting to recurring service interruptions and reduced customer satisfaction.

This study addresses the need for a proactive approach by implementing a GIS-based fault prediction approach that applies predictive modelling to integrate fault records with spatial and environmental data to identify fault risk areas, allowing the enhancement of network monitoring and improvement in service reliability.

1.4 Research Objectives

This research focuses on achieving the following objectives set.

1.4.1 Main Objective

To predict the likelihood of fault occurrence and map the fault-risk areas associated with TTCL's fiber-optic network in Dodoma Municipal, to support proactive maintenance and improve service reliability.

1.4.2 Specific Objectives

i. To identify and analyse the spatial and environmental factors that contribute to network faults in the infrastructure of TTCL.
ii. To estimate the likelihood of fault occurrence and map the high-risk areas within TTCL's fiber-optic network using predictive modelling.
iii. To validate the predictive performance of the predictive model.

1.5 Research Questions

i. What spatial and environmental risk factors contribute to network faults in TTCL's infrastructure?
ii. How can predictive modelling be applied to estimate the likelihood of network faults and map high-risk areas?
iii. How effective is the predictive model in predicting the fault-risk areas?

1.6 Significance of the study

This research is important as it highlights the challenge that TTCL faces in attending or estimating the possible locations where fault could occur and its management within its network infrastructure. By implementing a GIS based predictive modelling approach, it will introduce a proactive strategy towards identifying where faults are mostly likely to occur which

will then reduce response times and improve customer satisfaction by improving reliability and improving maintenance planning. The findings will help TTCL enhance its operational efficiency, retain customers, and strengthen its competitive position in the telecommunications market. Generally, the research will broaden the application of GIS in infrastructure management of which will provide a foundation for similar studies in other sectors of interest.

1.7 Beneficiaries

i. **TTCL**: By improving the estimation and identification of fault occurrences at early times, it will bring a positive impact in enhancing the reliability of the services of which facilitates satisfaction to its customers and boost up its revenues for which will be used to address areas of improvement.

ii. **Customer**: While the most important ends are the customers themselves who are in need of the service, a reduction in service disruptions will help retain their trust in the service they provide and while communication is the key aspect, such experiences can be improved.

iii. **The Telecommunication Industry**: By achieving insights from this study, it can be addressed and applied or reflected to similar challenges that are being faced or foreseen upon in the other organizations

iv. **Academic & Researcher Institutions**: This research will contribute to the continuing development body regarding the spread of the knowledge on Geographic Information System in terms of its applications in the management of infrastructure.

1.8 Scope and Limitations

This study focuses on implementing a GIS-based fault prediction approach for TTCL's telecommunication infrastructure in Dodoma Municipal by integrating the spatial and environmental data such as elevation, infrastructure and rainfall data along with fault records to identify the high-risk areas. The approach therefore utilizes both aerial and underground fiber-optic cable fault reports by utilizing tools like R Studio, QGIS as well as Arc GIS for spatial analysis, predictive modelling and validation. However, this research is limited by restricted access to TTCL's internal operational data, inconsistencies in its fault reporting, and the resolution of environmental datasets. It also excludes real-time detection capabilities due to resource constraints and is geographically limited to Dodoma Municipal, meaning that the results may not be directly generalizable to other regions without further model adjustments.

1.9 Dissertation Organization

This dissertation is organized into six chapters of which each addresses a key aspect of the research process starting from conceptualization towards the implementation of the predictive modelling approach and its application. The structure is therefore designed to provide a logical flow of ideas, methodologies, results, discussions and interpretations. In brief the parts will be as follows;

Chapter One: This chapter introduces the research by presenting the background, problem statement, research objectives and questions, significance, scope and limitations.

Chapter Two: Provides a comprehensive review of related literature, focusing on telecommunication fault prediction, GIS applications in infrastructure monitoring, and environmental factors influencing network reliability. It also highlights knowledge gaps this study seeks to fill.

Chapter Three: This chapter will outline the research methodology, detailing the data sources, data processing techniques, GIS tools, and implementation framework of the predictive modelling approach. It also includes the validation procedures and fault mapping approaches used in the study.

Chapter Four: Presents the results of the spatial analysis and GIS-based fault predictive modelling implementation. It discusses the identified fault risk zones, visualized outputs (e.g., fault risk maps), and the interpretation of key findings in relation to the research objectives.

Chapter Five: Introduces major discussions from the results pointed out in chapter four obtained in the dissertation by relating with what the literatures have inferred, as well as discussing their implications for TTCL and the broader telecommunications sector.

Chapter Six: The chapter finalises or concludes the dissertation by summarizing the key findings and assessing how the objectives were met. It therefore offers practical recommendations for improving TTCL's fault management using GIS and lastly, highlights research limitations and proposes areas for future investigation.

CHAPTER TWO
LITERATURE REVIEW

2.1 Overview

This chapter examines the existing research on telecommunication network fault approaches to manage and deal with it by emphasizing the role of Geographic Information Systems in infrastructure management. It will therefore look into the types of network faults being engaged in this research, the limitations existing by incorporating the use of traditional reactive approach method, and how GIS can help identify and manage areas with the maximum likelihood of faults occurrences. By analysing previous studies and insights from various experts, this chapter will highlight gaps that this study aims to address, particularly, the underuse of GIS in early fault resolution within Tanzania's telecommunication sector.

2.2 Tanzania Telecommunications Corporation Limited

The Tanzania Telecommunications Corporation Limited was established in 1993 under the Tanzania Telecommunications Act of 1993. It arose from the restructuring of the former Tanzania Posts and Telecommunications Corporation (TPTC), which previously had involved itself in managing the postal, telegraph, and telephone services in Tanzania. The operations of TTCL had started to commence on January 1, 1994, under the government's decision to liberalize the telecommunications sector.

Initially, TTCL was wholly owned by the Tanzanian government, but on February 23, 2001, it went through partial privatization embedded with a consortium from the Netherlands and Germany acquiring 35% of its shares. But, with all these on June 23, 2016, the Tanzanian government fully repossessed TTCL, making it 100% state-owned once again. The transition was formalized under the Tanzania Telecommunication Corporation Act No. 12 of 2017, which took effect on February 1, 2018. (United Republic of Tanzania, 2017)

TTCL remains a key player in Tanzania's telecommunications sector, providing fixed-line, mobile, broadband, and ICT services across the country. Its historical evolution reflects the broader shifts in Tanzania's telecom industry, balancing public ownership with market-driven reforms.

2.3 Fault Prediction Approaches

Predicting faults in modern systems, be it in telecommunications, environmental monitoring, or infrastructure, has become a critical area of research that associates engineering with data science. Fault prediction primarily aims at anticipating system failures before they occur by using patterns which are observed in historical or real-time data. In the telecommunications context, experts like Thakre and Kolhe (2014) emphasized that early fault prediction in access networks can significantly reduce downtime and improve maintenance. Ghorbanzadeh et al. (2019) demonstrated the need of incorporating spatial variables into machine learning models within a GIS environment to enhance predictive accuracy which is a concept that is widely used in landslide susceptibility studies but adaptable to telecom infrastructure. Moreover, researchers like Hong et al. (2016) and Khosravi et al. (2020) advocate that prediction models combining environmental variables and geospatial techniques perform better than traditional statistical methods. These insights therefore show a strong pattern across different fields, starting with the environment reaching into infrastructure, and toward proactive forecasting which is enabled by the integration of GIS and advanced analytics.

2.4 Modelling Approaches in Fault Prediction: GLM and ROC Analysis

Fault prediction modelling involves carefully building the equations or algorithms that describe how different factors relate and affect outcomes. Basically, a model acts as a mathematical representation of reality, allowing us to simulate possible scenarios within a system based on certain conditions. Researchers such as Naghibi and Pourghasemi have showed that; predictive modelling in fault detection, whether in groundwater mapping or infrastructure systems, benefits from integrating environmental and spatial data. One of the most widely used statistical techniques in fault prediction is the Generalized Linear Model (GLM), which extends the traditional linear regression framework to allow for non-normal distributions and includes a link function to relate the mean of the response variable to a linear combination of predictors. As defined by McCullagh and Nelder (1989), a GLM is "a class of models in which the distribution of the response variable belongs to the exponential family, and the expected value is connected to the linear predictors through a specified link function." This makes GLM particularly effective for binary classification tasks such as fault presence or absence. To evaluate the performance of such predictive models, Receiver Operating Characteristic (ROC) analysis is commonly applied. As described by Fawcett (2006), "ROC analysis provides a comprehensive visual and quantitative method for assessing the performance of binary classifiers by plotting the true positive rate against the false positive rate at different thresholds,

with the Area Under the Curve (AUC) serving as a summary measure of model accuracy." The combined use of GLM for fault likelihood estimation and ROC-AUC for model validation, establishes a dependable, transparent, and scientifically validated approach to spatial fault modelling.

2.5 Network Faults in Telecommunication Infrastructure

Network faults in telecommunication infrastructure refer to disruptions in signal transmission, often caused by physical, electrical, or environmental factors. These faults are broadly classified by type, such as open circuits, short circuits, or signal attenuation, and by category, talking about the environment where it's used, namely aerial or underground systems. Aerial faults typically result from environmental exposure, including wind-induced vibrations, temperature fluctuations, lightning strikes, and physical interference from trees or vehicles, making them more visible yet susceptible to external stressors (Thakre & Kolhe, 2014). In contrast, underground faults are harder to detect due to their disguise nature although they are primarily driven by factors such as moisture infiltration, ground displacement, rodent damage, and insulation fatigue (Alagumariappan et al., 2019). High-altitude regions intensify aerial cable challenges, increasing weaknesses due to cold temperatures and wind stress (Xu & Yuan, 2023). Recent advances in fault detection support machine learning and neural networks for real-time prediction and localization, outperforming traditional methods like time-domain reflectometry (Zhang et al., 2023; Zhou et al., 2020). Understanding the interaction between fault types and deployment categories is important for designing resilient telecommunication systems and optimizing maintenance strategies.

2.6 Geographical Information System

Geographic Information Systems are powerful tools used to capture, store, manipulate, analyse, manage, and visualize spatial or geographic data. According to Longley et al. (2015), GIS enables users to understand patterns, relationships, and geographic contexts by linking location-based data with attribute information. This integration of spatial and non-spatial data facilitates thorough spatial analysis and decision-making across various fields.

In the telecommunications sector, GIS plays an important role in infrastructure planning, network management, fault detection, and service optimization. As described by Cova (1999), GIS supports spatial analysis by mapping network layouts such as Fiber optic cables, base stations, and customer locations, enabling providers to plan and optimize service delivery more

efficiently. GIS helps in identifying service gaps, predicting high-risk areas for network faults, and planning expansions with minimal costs and maximum coverage.

GIS integration with other datasets, such as elevation, weather, land use, and population data; enhances the capacity of telecom companies to assess risk factors affecting network reliability. For instance, Zhou et al. (2020) demonstrated that overlaying environmental factors on network layers helps identify regions prone to outages due to weather or terrain challenges. Therefore, the use of GIS ensures a reliable delivery of important resources required for effective network deployment, monitoring, and fault management. Nevertheless, GIS contributes to improved service quality, customer satisfaction, and operational efficiency.

2.7 Spatial and Environmental Risk factors

Spatial and environmental risks refer to location-based and natural factors that increase the vulnerability of infrastructure and services to failure or damage. Spatial risks include human-made features such as proximity to roads, buildings, and land use patterns that can physically or functionally interfere with infrastructure, while environmental risks involve natural conditions like rainfall, temperature extremes, soil moisture, and vegetation cover that can affect performance or durability. In urban settings, these risks are often interconnected, such as when informal settlements are built close to infrastructure, they intensify stress during floods or extreme heat events. The United Nations Office for Disaster Risk Reduction (2020) emphasizes the importance of including both types of risks in disaster and infrastructure planning frameworks, especially as climate change intensifies fluctuations in environmental conditions. GIS provides effective means to map and analyse these risk factors spatially, allowing planners to identify high-risk areas, model fault probabilities, and allocate resources more efficiently. Freely accessible tools and data from OpenStreetMap (2023) and FEMA (2022) support this process, helping conduct detailed evaluations at a low expense of spatial and environmental vulnerabilities, thereby reinforcing the role of open geospatial technologies in building resilient infrastructure systems.

2.8 Spatial Analysis

Spatial analysis refers to the set of techniques used to examine spatial data for patterns, relationships, and structures across geographic space. It's different from basic spatial summaries because it aims to explain how spatial factors vary and interact and not just listing or describing them. According to Fotheringham and Rogerson (1994), spatial analysis covers

both exploratory and confirmatory methods and is increasingly integrated with Geographic Information Systems to enhance visualization, spatial querying, and pattern detection.

Spatial analysis in the telecommunications sector, particularly for companies like Tanzania Telecommunications Company Limited, is important for optimizing network infrastructure, improving service delivery, and enhancing customer satisfaction. By integrating Geographic Information Systems, telecom operators can visualize and analyse spatial data to determine ideal locations for cell towers and Fiber-optic routes, assess service coverage using viewshed analysis, and manage assets more effectively. In TTCL's case, spatial segmentation revealed key urban markets and demographic patterns influencing service preferences, enabling more targeted and competitive strategies (Semboko, 2015). Furthermore, GIS supports proactive maintenance and customer service improvements by identifying network weak points and planning upgrades accordingly (Cova, 1999).

2.8.1 Spatial Analysis Techniques

Spatial analysis techniques offer a wide range of tools for exploring how things vary and interact across space so as to examine the spatial relationships, patterns, and structures within geographic data. In the context of telecommunication infrastructure, these techniques are important in identifying and analysing environmental factors that contribute to network faults.

One significant study by Cai (2002) used GIS to assess telecommunications infrastructure, highlighting how spatial analysis can reveal areas susceptible to service disruptions due to environmental conditions. Similarly, Khosravi et al. (2020) applied spatial and statistical techniques to identify weak zones and predict infrastructure vulnerability using environmental and topographic variables. Therefore, there has been an incorporation of several analysis conducted in this research comprising of both buffer analysis, terrain analysis, and correlation analysis of which where critically explored.

- Correlation Analysis

  Correlation analysis is a statistical method used to examine the relationships between two or more variables, particularly to determine the strength and direction of their association. In the context of telecommunication infrastructure, correlation analysis has been applied to identify environmental factors that influence network stability and fault occurrences. For instance, studies have shown that with the application of correlation analysis, specifically in network optimization and maintenance, correlating historical fault records and data (be real-time data or available from external sources) of relevant

use such as datasets about environment, can help in minimization of disruption and thus prioritizing maintenance scheduling (NumberAnalytics, 2021). Even if it's not always used in complex models, correlation analysis is a crucial first step that helps researchers figure out which factors whether it is environmental or infrastructure, should carry more weight in later spatial modelling or risk prediction.

- Buffer Analysis

  It is a fundamental GIS operation used to determine zones of influence around geographic features, such as telecommunication infrastructure. It helps in assessing proximity-based risks, like potential interference from nearby structures or environmental hazards. The technique has been utilized in various applications including telecommunications (de Smith, Goodchild, & Longley, 2018)

  In assessing proximity relationships between telecommunication infrastructure, such as Fiber optic cables and buildings, and potential environmental or anthropogenic risk factors. In the context of TTCL, buffer zones (e.g., 10–50 meters around Fiber lines and 5–15 meters around buildings) can be used to evaluate areas where network faults are more likely due to nearby construction, vegetation, or unauthorized access. Cai (2002) demonstrated how buffer analysis helps spatially relate infrastructure with service demand and risk zones, while Akdemir et al. (2016) applied GIS buffers in a Fiber network monitoring system to detect maintenance risks.

- Terrain Analysis

  Terrain analysis evaluates the physical characteristics of the Earth's surface, particularly elevation, slope, and aspect, which are key determinants in the placement and reliability of telecommunication infrastructure. Steep slopes, for instance, can complicate underground cable installation and increase susceptibility to erosion or landslides. Mappiakka (2015) showed that in hilly areas, signal strength and reliability were directly influenced by terrain configuration. In GIS, terrain analysis is often used in combination with other layers (e.g., rainfall and land cover) to better understand environmental impacts on network performance.

2.9 Accuracy Assessment

Accuracy assessment refers to the process of evaluating the correctness and reliability of data or outputs generated from a model, system, or process. In Geographic Information Systems and Remote Sensing for example, it commonly involves comparing classified data (like land cover maps) with ground truth or reference data to measure the level of agreement. Metrics

such as overall accuracy, user's accuracy, producer's accuracy, and the Kappa coefficient are typically used (Congalton & Green, 2019).

In the telecommunication sector, accuracy assessment plays a critical role in ensuring the precision of spatial data used in planning and monitoring network infrastructure. For instance, when developing or implementing GIS-based models for fault mapping in aerial or underground fiber-optic cable networks, an accuracy assessment is conducted to evaluate how well the model predicts fault-prone areas or network outages. High accuracy indicates that the system can reliably guide maintenance operations, optimize resource allocation, and improve service delivery.

Upon the evaluation of spatial predictive models, particularly in spatial analyses involving binary classification tasks such as fault presence, they normally or typically involve comparing predicted outcomes to observed data being used. This process ensures the model's reliability and generalizability. Logistic regression has been widely used in many cases, for example, in spatial modelling of landslides and ecological risks, as noted by Elith et al. (2006) and Naghibi & Pourghasemi (2015). In such studies, spatial environmental variables like elevation, rainfall, and temperature extremes are extracted at known event locations (e.g., landslide sites or fault reports) and compared against randomly sampled background points to train the model. After model fitting, predictions are evaluated using metrics like the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC), with the pROC package in R Studio commonly used for this purpose (Fawcett, 2006). The methodology followed in this dissertation is inspired by these established approaches, adapted to the context of telecommunication infrastructure fault prediction in Dodoma Municipal.

CHAPTER THREE
METHODOLOGY

3.1 Overview

In this part of the methodology; the methods, techniques, and data collected in the study are well explained. It therefore describes all the processes from data collection and manipulation towards the implementation of the predictive model for the estimation of fault occurrence and mappings of the risk areas affecting TTCL.

3.2 Description of the study area

Dodoma is the capital city of Tanzania that is located at the central part of the country (see Figure 3.1). It is an administrative area as well as an infrastructural hub that experiences rapid urbanization driven by governmental, residential, and commercial development. The region has become a focal point for infrastructure investment, including the expansion of telecommunication services. Dodoma lies approximately between 6.00°S and 6.20°S latitude, and from 35.70°E to 36.00°E longitude (United Republic of Tanzania, 2022). The area is characterized by a semi-arid climate, meandering terrain, and seasonal rainfall patterns, which influence infrastructure reliability and susceptibility to environmental stress. Despite the company's service provision and operations through a wide range of infrastructures including Fiber Optic Cables, Mobile Towers, and a Fixed Network; TTCL has faced challenges in maintaining uninterrupted service delivery due to terrain variability and environmental conditions. These factors, in combination to ongoing construction activities, increases the risk of network faults hence making Dodoma a relevant case study area for GIS-based fault prediction.

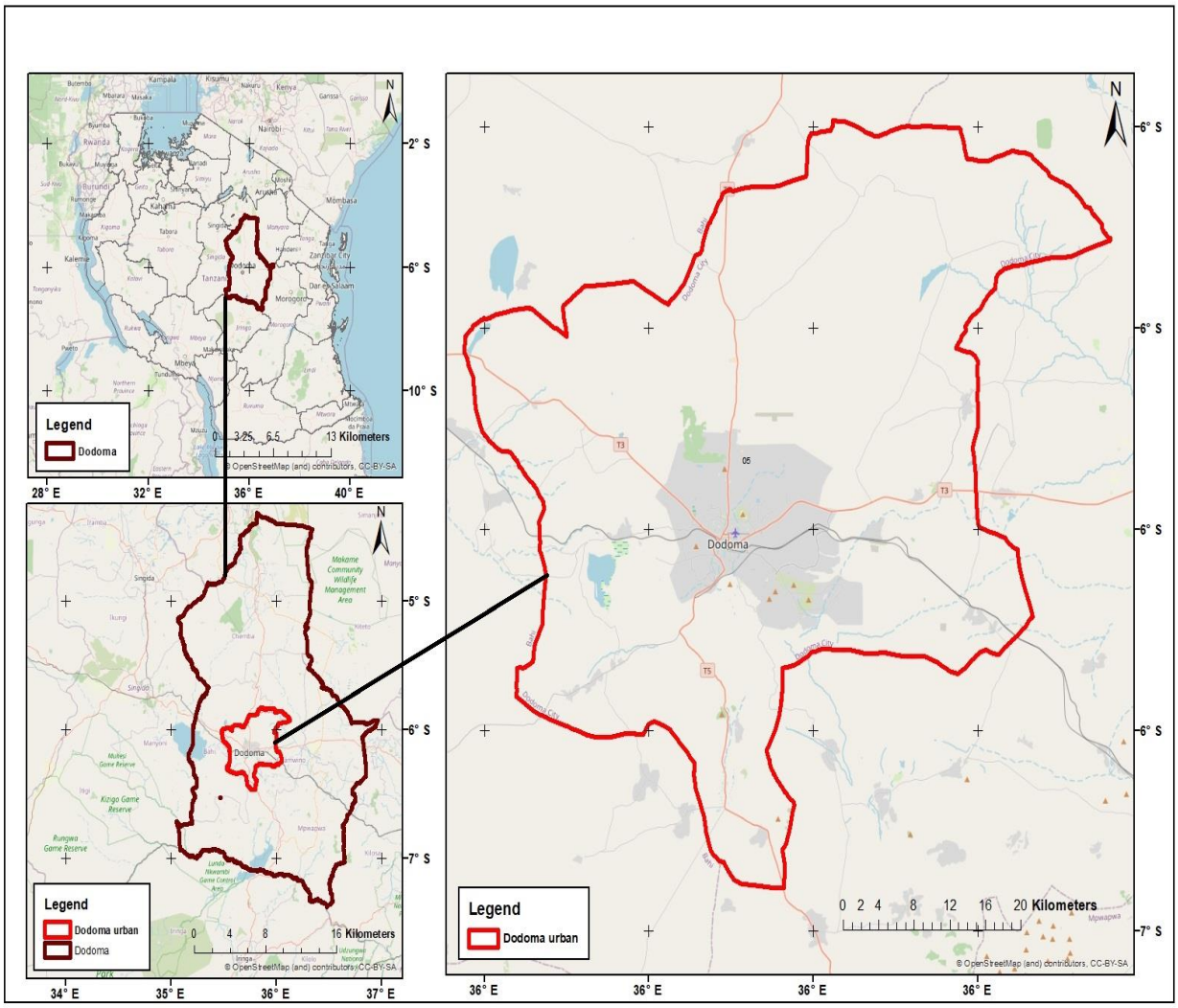

Figure 3.1: Dodoma Urban Location Map

3.3 Methodology Workflow

The following methodological workflow, as shown by Figure 3.2, details what steps are to be engaged for this research to meet its intended objectives and how it has enabled the research to work upon obtaining the results.

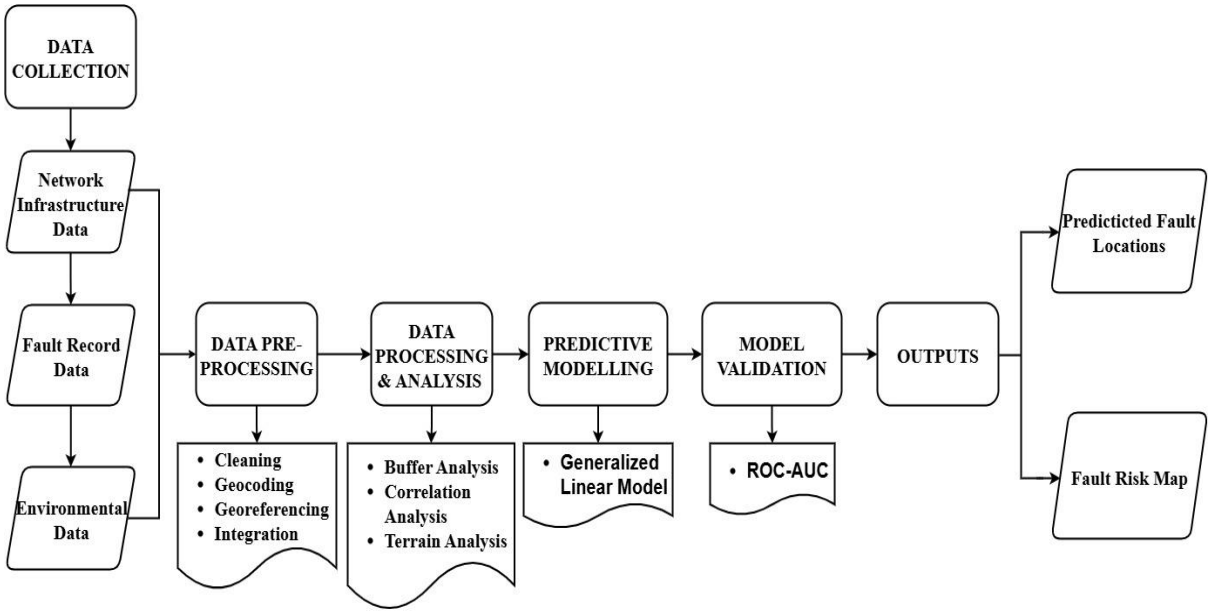

Figure 3.2: Research Methodology Workflow

3.4 Data Collection

In this process, most of the data had been collected, as tabulated in Table 3.1, with respect to what the literature review has pointed out to be effective in engagement for the success of this research. The literature part suggested the engagement and the need for acquiring datasets such as Fault Record Data, Environmental Data (Soil Moisture Data, Precipitation Data, Temperature Data, Elevation Data), and Network Infrastructure Data (Aerial and Underground cable information).

Table 3.1: Research Datasets

S/N	DATASET	SOURCE	FORMAT	SPATIAL RESOLUTION	USAGE
01.	Network Infrastructure Data	TTCL	GeoJSON	<3m	Identifying spatial layout and types of cables for mapping
02.	Fault Records Data	TTCL	CSV		Analysing fault occurrences
03.	Elevation Data	USGS (SRTM)	GeoTIFF	30m	Assessing terrain influence on fault distribution
04.	Rainfall Data	Terra Climate	NetCDF	~ 4km	Evaluating correlation of

S/N	DATASET	SOURCE	FORMAT	SPATIAL RESOLUTION	USAGE
					rainfall intensity & fault occurrence
05.	Soil Moisture Data	Terra Climate	NetCDF	~ 4km	Evaluating correlation of soil moisture influence to fault occurrence
06.	Temperature Data	Terra Climate	NetCDF	~ 4km	Assess the impact of thermal stress on cable reliability

3.5 Data Pre-processing

- **Fault Data**

The obtained data of Fault records, as shown in Table 3.2, was pre-processed by undergoing a series of data cleaning to remove data with no values (NA) and geocoding for the purpose of providing geographical coordinates corresponding to a location of fault and in our case, converting an address into geographical coordinates. The following is the summary. The TTCL firm, under the Access Network Office, issued this data and they had to filter only those faults for the corresponding region of interest and these faults are recorded with respect to the government calendar of finance (July-June) each financial year.

Table 3.2: Sample of Raw TTCL Fault Records

F1220 | CRDB BANK

	B	C	D	E	F	G	H	
1	REGION_NAM	Area	oduct Na	Access No.	Customer Name	tomer T	omer Ad	
12	DODOMA	Dodoma	FIXED DA	BB262323384		Individua	DODOMA	MR
14	DODOMA	Dodoma	FIXED DA	BB862350181	KISSA VIVIAN KILINDU KILINDU	Individua	AREA E N	KISSA VI
23	DODOMA	Dodoma	FIXED DA	BB862300100	KIMBE AUTO GARAGE	Individual		Mr.
27	DODOMA	Dodoma	FIXED DA	BB862322890	SULEIMANI HAMADI	Individual		suleimar
28	DODOMA	Dodoma	FIXED DA	BB262321260	BRITAM INSURANCE (TANZANIA) LIMITED	Corporat	DAR ES S	255 767
33	DODOMA	Mpwapw	FIXED DA	BB862380017	JUDICIARY OF TANZANIA	Corporat	BOX 900	witness
37	DODOMA	Kondoa	FIXED DA	BB262360356	TANZANIA ELECTRIC COMPANY LTD	Corporat	Dar es Sa	Mob#06
55	DODOMA	Dodoma	FIXED DA	BB862352236	KHALID THABIT	Individual		KHALID
60	DODOMA	Dodoma	FIXED DA	BB262329022	TANZANIA COMMERCIAL BANK PLC	Corporat	DAR ES S	Henry S
62	DODOMA	Dodoma	FIXED DA	BB262351199	WORLD BANK TANZANIA	Corporat	PO BOX 2	chirspiar
66	DODOMA	Dodoma	FIXED DA	BB862350184	HUNGU INNOCENT JAMES	Individua	AREA D(K	HUNGU
67	DODOMA	Dodoma	FIXED DA	BB262380000	CRDB BANK	Corporat	P.O.BOX	BUYEGI
68	DODOMA	Dodoma	FIXED VC	255262321345	HELVETAS HELVETAS	Individua	P.O.BOX	MR

DODOMA

The fault data had the following columns: TT No., REGION_NAME, Area, Product Name, Access No., Customer Name, Customer Type, Customer Address, Contact Person, Contact No., TT Title, Status, Report Date, Duration Hours, Staff Station, and Staff Name. Some columns (Customer Name, Customer Address) had empty fills in one of their rows indicating no record was issued. Geocoding was performed using R studio under the package named, "tidygeocoder." The steps towards achieving the geocoding process were as follows: Step 1: Loading the CSV file containing fault records, Step 2: Renaming the columns for convenience, Step 3: Cleaning the addresses and tag PO Boxes, Step 4: Geocoding process under the method "osm", Step 5: Converting the data to spatial object in WGS 84, and the last step (Step 6) was to save the result after the geocoding process was a success.

The Geocoded Fault Data had now comprised of Latitude and Longitude Columns as an addition simply signifying location of fault to which had been reported by the customers receiving the service of network. Table 3.3, shows the newly appearance of geocoded fault data with a change of column names and additional columns of Latitude and Longitude.

Table 3.3: TTCL Fault Records After Pre-processing

A1 | TT No.

	F	G	H	I	J	K	L	M	N	O	P	Q
1	Customer I	Customer `	address	Contact Pe	Contact N	TT Title	Status	Report Date	Duration H	full_addre	latitude	longitude
2	HUSEN KO	Individual	NA	HUSEN KO	6.8E+08	Unissued \	COMPLETE	##############	18.52	NA, DODO	-6.17912	35.7468
3	FATUMA S	Individual	NA	SALIM	NIL	Unissued \	COMPLETE	##############	5.72	NA, DODO	-6.17912	35.7468
4	SELEMAN	Individual	Dodoma	SELEMAN	NIL	Unissued \	COMPLETE	##############	52.38	Dodoma, D	-6.17912	35.7468
5	BONIPHAC	Individual	Mpakani K	BONIPHAC	7E+08	Unsuccess	COMPLETE	##############	81.37	Mpakani K	NA	NA
6	SAUMU M	Individual	NA	SAUMU M	7.4E+08	Unissued \	COMPLETE	##############	40.71	NA, DODO	-6.17912	35.7468
7	KENYATA J	Individual	NA	KENYATA J	7.1E+08	PIN Reset	COMPLETE	##############	4.12	NA, DODO	-6.17912	35.7468
8	SAIDI ATH	Individual	NA	SAID MHIN	6.2E+08	PIN Reset	COMPLETE	##############	1.8	NA, DODO	-6.17912	35.7468
9	NATHANIE	Individual	Gwandi	DANI	6.8E+08	Unsuccess	COMPLETE	##############	61.3	Gwandi, D	-5.3034	35.8219
10	PAUL ELIA	Individual	Mkapa Sha	PAUL ELIA	7.4E+08	PIN Reset	COMPLETE	##############	2.6	Mkapa Sha	NA	NA
11	SHABANI C	Individual	Dodoma	SHABANI	7.7E+08	PIN Reset	COMPLETE	##############	2.23	Dodoma, D	-6.17912	35.7468
12	NA	Individual	Dodoma	MR	7.4E+08	On & off	COMPLETE	##############	95.23	Dodoma, D	-6.17912	35.7468
13	DINNA EM	Individual	NA	DINNA EM	NIL	PIN Reset	COMPLETE	##############	1.91	NA, DODO	-6.17912	35.7468
14	KISSA VIVI	Individual	Area E Nea	KISSA VIVI	7.7E+08	Internet fa	COMPLETE	##############	22.44	Area E Nea	NA	NA

After the geocoding process was successful, some columns such as; Customer Name, Address, Contact No., Full Address, Latitude, and Longitude. Had NA, and NILL values in some of their rows which had to remove to remain with clean data to use. The visualization of the fault data after pre-processing in QGIS are shown in Figure 3.3.

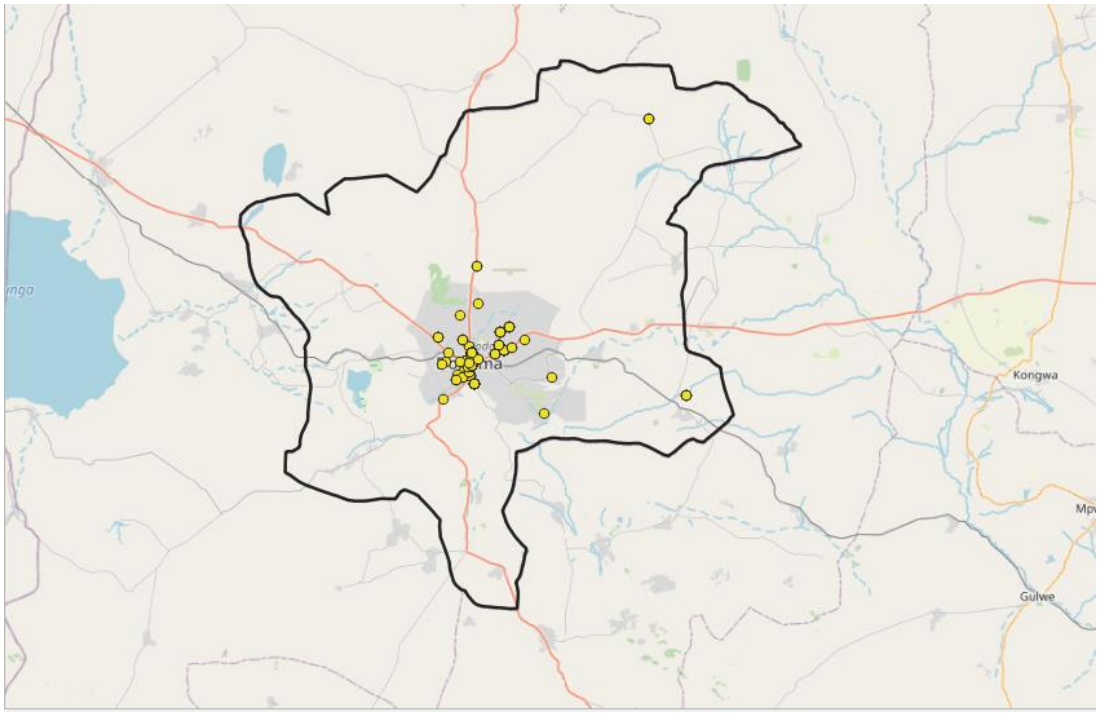

Figure 3.3: Visualization of Geocoded Fault Data in QGIS Software

- **Elevation Data**

Data for elevation was initially obtained through USGS (SRTM) source (https://urs.earthdata.nasa.gov) and in this case, was directly downloaded through QGIS software under the SRTM Downloader Plugin. In order to access the data, the login credentials were needed and so the requested details were filled and the results are shown in Figure 3.4.

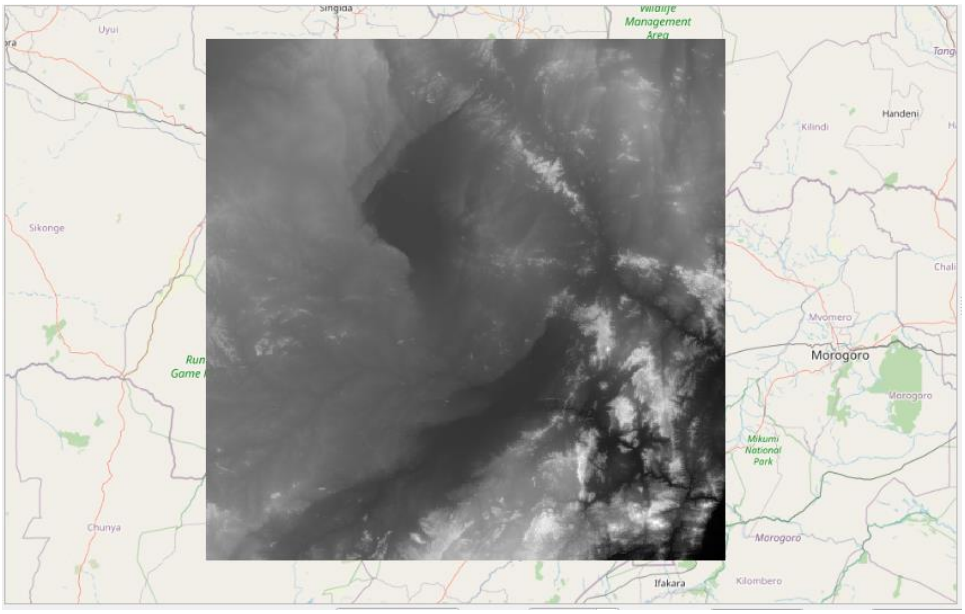

Figure 3.4: Elevation Raster

The pre-processing of such data involved clipping the area of interest through the use of clipping tool available in the QGIS software accessible in the Raster Layer > Extraction > Clip Raster by Mask Layer. The results of the clipping process are shown in Figure 3.5.

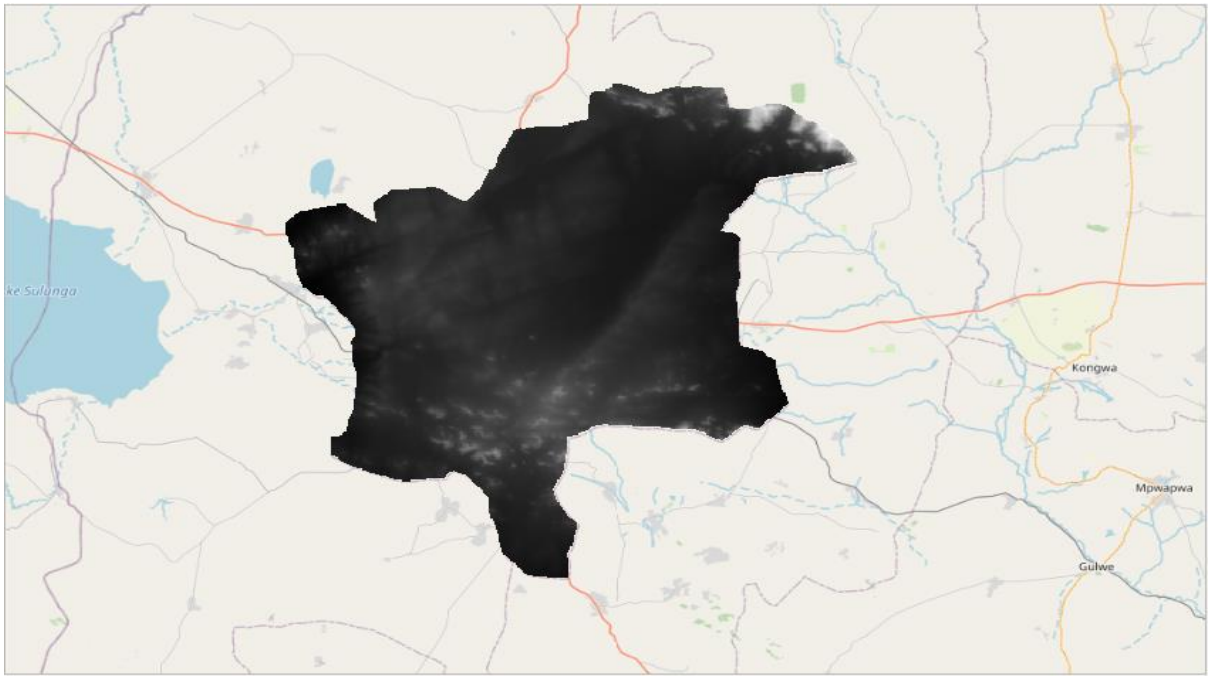

Figure 3.5: Clipped Elevation Raster of Dodoma Urban

- **Network Infrastructure Data**

Since it is important to map the physical components and identify the vulnerable areas, it was necessary to identify the location of important infrastructure which are, aerial and underground cables as well as manholes. The pre-processing part had involved the process of identifying the locations of these network infrastructure data from a general dataset obtained from TTCL. The general dataset provided by TTCL was a QGIS Database that showed all the available locations and details about their infrastructural mappings. Figure 3.6 shows how the generalized dataset of network infrastructure provided looks like.

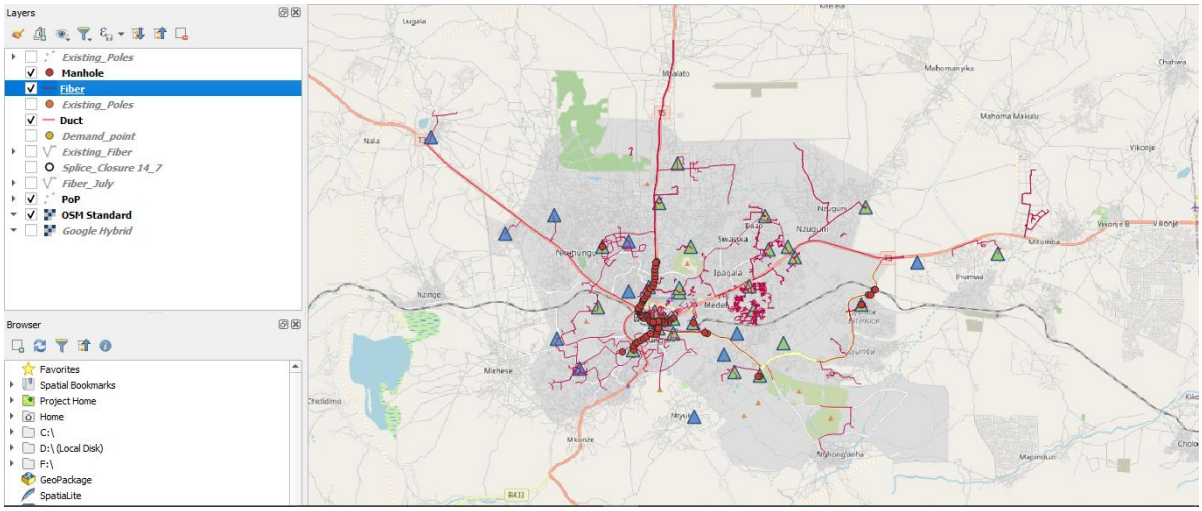

Figure 3.6: Network Infrastructure Layout

From the generalized dataset, in order to identify the aerial and underground fiber-optic cables, the attribute table of Fiber was opened and the required attributes from the field of Placement Type were selected: Aerial and Underground Fiber Cables. This is observed from Figure 3.7 showing the selection process.

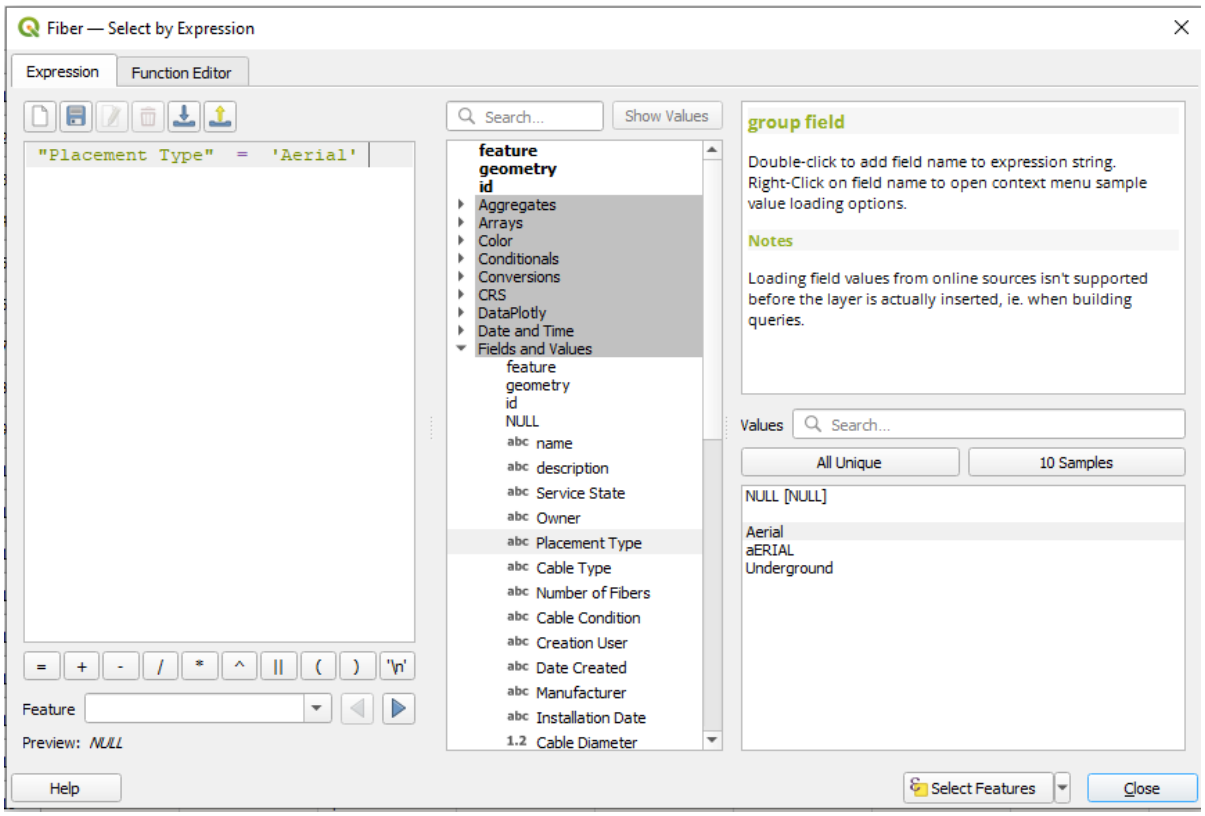

Figure 3.7: Selecting by Expression of Placement Types from the Attribute Table of Fiber

In a nutshell, the spatial and non-spatial datasets were sourced from TTCL, open data repositories such as OSM, and global climate datasets such as TerraClimate. All spatial layers were transformed into the EPSG: 32736 (WGS 84 / UTM Zone 36S) coordinate reference system to ensure consistency in spatial overlay and analysis. This re-projection was and is important to avoid spatial mismatches during raster-vector operations conducted for the attainment of the outlined objectives.

3.6 Data Processing and analysis

This study employed an integrated spatial-statistical approach using R and GIS tools to analyse environmental and spatial factors influencing TTCL network faults in Dodoma. The analysis involved the following core steps:

- **Predictive Modelling and Model Validation**

A Generalized Linear Model (GLM) with a binomial family was applied to perform the predictive analysis so as to predict the probability of fault occurrence. The logistic model is important for binary classification and has been widely applied in fault and hazard prediction studies. The model formula included: (fault_presence ~ elevation + rainfall + soil_moisture + tmax + tmin, family = binomial, data = data_all)

The model estimates the influence of both environmental and built-environment predictors on the likelihood of fault occurrence. The inclusion of natural and anthropogenic variables conforms with established multi-criteria fault and hazard assessment practices cited in existing literature. Evaluation in terms of performance was assessed and validated using the AUC of the ROC based on the ROC–AUC validation which is common when evaluation, for this case, binary classification models; computed through the pROC package. An AUC score above 0.7 indicates acceptable model discrimination (Fawcett, 2006).

- **Risk Map Generation and Visualization**

To spatialize fault likelihood across Dodoma Municipal, raster layers of the predictors were cropped and resampled to match the resolution of the elevation raster, ensuring consistency. A logistic prediction surface was then generated using predict () function from the GLM model, resulting in a raster where each pixel contains the predicted fault probability. The resulting raster was masked using the Dodoma urban shapefile to restrict analysis to the area of interest. The raster was saved as a GeoTIFF for further use and integration into visualization applications.

From the predicted risk map generated, the high-risk points which had the most, top five percent hotspots, were extracted and filtered to a manageable sample size (n=100) using as.data.frame() function and random sampling. These points represent areas where proactive maintenance or monitoring would be most effective. In addition, all layers were reprojected to EPSG:32736 – WGS 84 / UTM Zone 36S for accurate spatial analysis and data visualization compatibility.

- **Correlation Analysis**

To evaluate the relationship between environmental factors and the spatial distribution of TTCL network faults in Dodoma urban, a correlation analysis was conducted. This statistical method assesses the strength and direction of linear relationships between variables using a correlation coefficient ranging from -1 to 1.

A variable selection of environmental variables suspected to influence fault occurrences were extracted from the combined model dataset, including elevation, rainfall, soil moisture, maximum temperature (Tmax), and minimum temperature (Tmin). A fault indicator was utilized signifying a binary variable (fault_presence) indicating whether a fault was recorded in a given location, enabling comparison with continuous environmental variables. Pearson correlation coefficient was also used as a correlation computational matrix so as to reflect insights upon the positive (direct) and negative (inverse) relationships. The visualization was performed in R studio were the two techniques: ggcorrplot and ggpairs from GGally package was used. The ggcorrplot was used to provide a correlation matrix heatmap, showing correlation strength and significance where as ggpairs was used to provide the pairwise scatter plots and distribution histograms as shown in Figure 3.8 for visual inspection of variable interactions, and the colour visualization of the presence or absence of faults.

Figure 3.8: Pairwise Correlation & Distribution of Faults Vs Variables

3.7 Accuracy Assessment

Accuracy assessment is a significant component in evaluating the reliability and predictive power of a spatial fault detection model. In this study, accuracy assessment was conducted using Receiver Operating Characteristic (ROC) analysis and the associated Area Under the Curve (AUC) metric, following the implementation of the logistic regression model.

3.7.1 ROC Curve and AUC Analysis

The ROC curve plots the true positive rate (sensitivity) against the false positive rate (1 - specificity) across various threshold levels. It provides a comprehensive view of the model's diagnostic ability, independent of a specific classification threshold (Fawcett, 2006).

In this research, the ROC curve was generated using the pROC package in R, based on the predicted probabilities from the GLM and is shown by Figure 3.9.

Figure 3.9: ROC Curve for Predictive Model

The Area Under the Curve (AUC) was calculated to quantitatively assess model performance. The AUC value ranges from 0.5 (no better than random) to 1 (perfect discrimination). The fault risk prediction model produced an AUC score of 0.979, indicating good model performance and suggesting that the model effectively distinguishes between areas with and without faults. We can observe this in Figure 3.10.

Figure 3.10: ROC Curve for Predictive Model with AUC

CHAPTER FOUR
RESULTS

4.1 Overview

This chapter presents the outcomes of the spatial and statistical analyses, demonstrating how environmental and spatial factors drive network faults in Dodoma's TTCL system. It therefore starts by summarizing the datasets and the geocoded fault points, then move into exploratory correlation assessments that pinpoint the strongest predictors. Next, we detail the logistic regression (GLM) results and evaluate the performance with ROC–AUC analysis. Furthermore, we showcase the GIS-based risk surface as produced from the predictive model so as to highlight the high-risk hotspots which may have a negative impact for both aerial and underground fiber-optic cables. Finally, we interpret these spatial patterns in the context of Dodoma's terrain and TTCL's operational priorities, drawing out implications for proactive maintenance and customer service reliability.

4.2 Predictive Model Summary and Validation

To assess the probability of faults occurring within TTCL's fiber-optic network, this study employed a predictive modelling approach which utilized a Generalized Linear Model formulated on the basis of the following key variables; elevation, rainfall, soil moisture, and temperature extremes (Tmax and Tmin). The variables were selected for their relevance in previous research and their documented impact on infrastructure performance. The use of logistic regression enabled the prediction of binary fault outcomes (Fault occurrence vs. Fault non-occurrence), enabling a framework for spatial risk estimation. By analysing the interaction between environmental variables and fault occurrences, this study assisted with the identification of infrastructure vulnerable zones and strengthens TTCL's preventative maintenance strategies.

The following is the Predictive Model Summary and Validation as conducted in R Studio with tabulations as observed by Table 4.1.

```
Call:
glm(formula = presence ~ elevation + rainfall + soil_moisture +
    tmax + tmin, family = binomial, data = data_all)
```

```
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -7.194e+02 5.361e+01 -13.419 < 2e-16 ***
elevation    1.396e-02 4.532e-03   3.080 0.00207 **
rainfall     2.508e+00 1.731e-01  14.491 < 2e-16 ***
soil_moisture 5.943e-02 3.732e-02   1.593 0.11125
tmax         2.902e+01 2.205e+00  13.157 < 2e-16 ***
tmin        -1.914e+01 1.849e+00 -10.352 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 1642.02 on 1459 degrees of freedom
Residual deviance:  393.97 on 1454 degrees of freedom
AIC: 405.97
Number of Fisher Scoring iterations: 8
```

Table 4.1: Predictive Model Summary

Variable	Estimate	Std. Error	Z value	P-value	Significance	Interpretation
(Intercept)	-719.40	53.61	-13.419	< 2e-16	***	Baseline log-odds when all predictors are zero.
Elevation	0.01396	0.00453	3.080	0.00207	**	A positive effect.
Rainfall	2.508	0.1731	14.491	< 2e-16	***	Strong positive effect.
Soil Moisture	0.05943	0.03732	1.593	0.11125	Not significant	Weak positive effect.

Variable	Estimate	Std. Error	Z value	P-value	Significance	Interpretation
Tmax (°C)	29.02	2.205	13.157	< 2e-16	***	Very strong positive effect.
Tmin (°C)	-19.14	1.849	-10.352	< 2e-16	***	Strong negative effect.

Significance Levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The Predictive model summary table, as tabulated in Table 4.1, presents the results from the predictive model used to estimate fault occurrence in TTCL's fiber-optic network. Key environmental factors like elevation, rainfall, temperature (tmax and tmin), and soil moisture were included. The results showed that rainfall and temperature extremes (both high and low) were the most significant predictors of network faults. Elevation also had a moderate effect, while soil moisture showed a weak influence and wasn't statistically significant.

Briefly, in other words we can argue from the above statement that, rainfall emerged as the most influential factor, suggesting a strong positive effect on fault occurrence. High maximum temperatures (Tmax) similarly increased fault likelihood, while lower minimum temperatures (Tmin) had a notable negative association which suggests that cooler conditions might mitigate fault risks. Elevation contributed a moderate but meaningful impact, whereas soil moisture did not reveal a clear correlation with faults in this model implemented in this study. Overall, the predictive reliability of the model was strongly established through a good AUC score of 0.979, indicating excellent performance in identifying fault-prone areas.

4.2.1 Predictive Model Fit Statistics

The predictive analysis fit statistics, as shown in Table 4.2, provides quantitative insights into how well the implemented predicted model (GLM) explains the variation in fault occurrence across the study area. The null deviance represents the fit of the implemented model with no predictors (only an intercept), serving as a baseline, while the residual deviance indicates the fit of the implemented model with the included predictors. A significant reduction from null deviance to residual deviance demonstrates that the model significantly improves upon the baseline.

The Akaike Information Criterion (AIC) is a comparative measure of the implemented model quality, balancing the model fit with complexity; lower AIC values indicate a better-fitting model with fewer unnecessary parameters. Lastly, the AUC value, derived from ROC analysis, assesses the predictive accuracy of the model, with a value of 0.979 indicating excellent ability to distinguish between fault-prone and non-fault-prone areas.

These statistics collectively confirm that the predictive analysis implemented in this study produces a strong and reliable estimates of fault occurrence probabilities, supporting the objectives of proactive maintenance planning within TTCL's fiber-optic network.

Table 4.2: Predictive Model Fit Statistics

Statistic	Value
Null Deviance	1642.02
Residual Deviance	393.97
AIC	405.97
Number of Observations	1460
Number of Iterations	8
AUC	0.979

The predictive modelling implemented in this study was conducted to estimate the probability of fault occurrence using key environmental predictors: elevation, rainfall, soil moisture, maximum temperature (tmax), and minimum temperature (tmin). The results revealed a strong relationship between most of these variables and the presence of faults. Among them, rainfall showed the highest impact with an estimated coefficient of 2.508 and a very high level of significance ($p < 2e\text{-}16$), indicating that areas with more rainfall are notably more prone to faults. Tmax (29.02) and Tmin (-19.14) were also highly significant, suggesting that thermal stress, both from extreme heat and cold, can critically influence cable integrity. Elevation was statistically significant ($p = 0.00207$), suggesting that areas with higher elevation are marginally more fault-prone, possibly due to pole instability and terrain stress. Soil moisture showed the least influence and was not statistically significant ($p = 0.11125$), though it remains environmentally relevant.

The predictive model showed a significant reduction in deviance, with the residual deviance dropping from 1642.02 to 393.97, and a low AIC value of 405.97, indicating a well-fitting model. Furthermore, the AUC from ROC analysis was 0.979, demonstrating excellent predictive performance in distinguishing fault-prone areas from stable ones. This predictive analysis therefore fulfills specific objectives two and three by effectively estimating fault likelihood and validating the accuracy of the implemented fault-risk model.

Therefore, these findings validate the model's statistical reliability and real-world applicability in identifying key environmental contributors to network faults. Hence, it establishes a useful framework for TTCL to adopt proactive maintenance strategies that are built in actual spatial and environmental insights.

4.3 Correlation Analysis

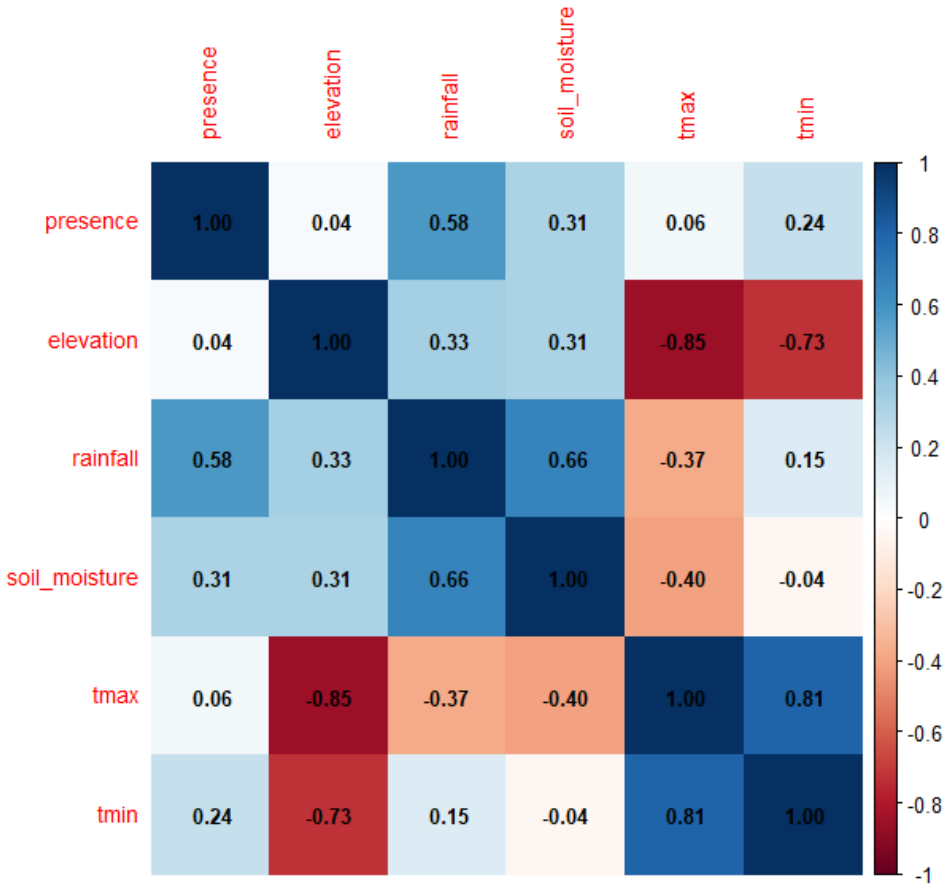

Figure 4.1: Correlation Matrix of Fault Presence Vs Environmental Risk Factors

The Pearson correlation matrix, as presented by Figure 4.1, was used to examine the strength and direction of relationships between fault presence and various environmental variables including elevation, rainfall, soil moisture, tmax, and tmin. The analysis revealed that rainfall

had the strongest positive correlation with fault presence (r = 0.58), suggesting that areas with higher precipitation are more prone and susceptible to network faults. This could be due to increased soil saturation or waterlogging, which tend to affect underground fiber-optic cables or pole-based aerial infrastructure. Soil moisture also showed a moderate positive correlation (r = 0.31), indicating that excess water in the ground may contribute to corrosion and structural instability around network components. Tmin (r = 0.24) had a weak but notable positive correlation, while elevation (r = 0.04) and tmax (r = 0.06) showed almost no correlation with fault presence. We can also note that, the matrix also showed a strong negative correlation between elevation and tmax (r = -0.85), and a strong positive relationship between tmax and tmin (r = 0.81), suggesting that elevation plays a role in moderating temperature, although not directly influencing faults.

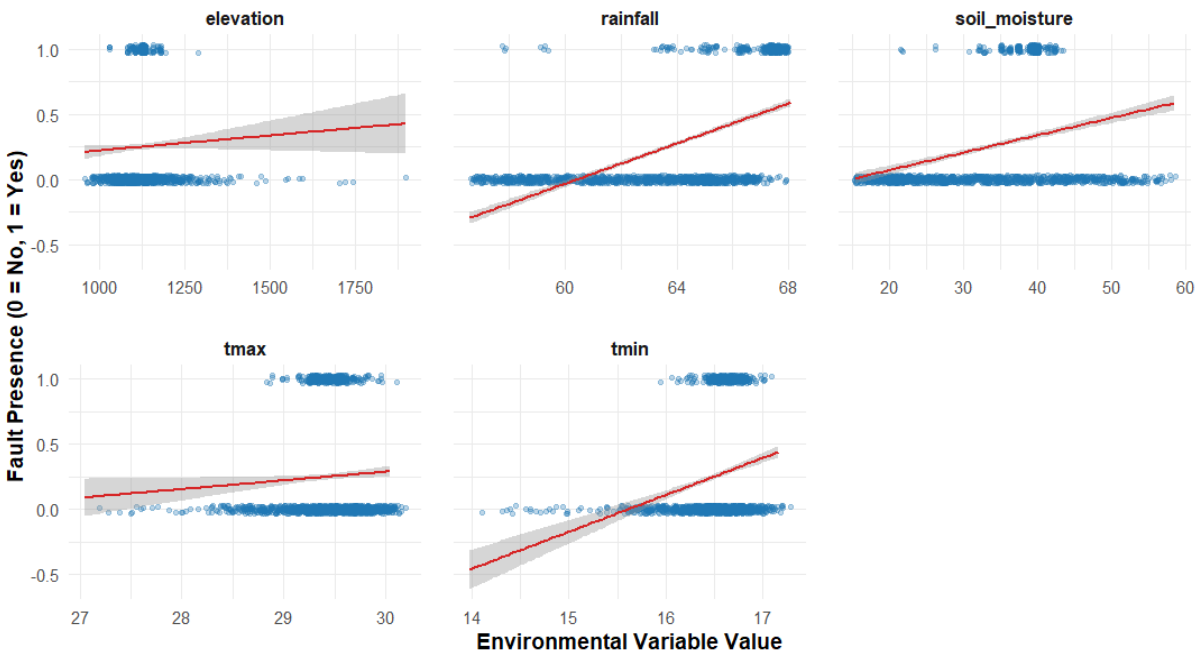

Figure 4.2: Scatterplots of Environmental Risk Factors Vs Fault Presence

The scatterplots, shown in Figure 4.2, provide a visual representation of how each environmental variable relates to fault presence. Rainfall shows a clearly visible upward trend, supporting the statistical results from the correlation matrix whereby, areas with higher rainfall consistently showed increased fault presence, reinforcing the idea that water is a major contributor to network failures. Soil moisture showed a similar, although weaker, positive trend. Tmin reveal slight upward variation, further suggesting some influence, especially in cooler regions where condensation might affect infrastructure. On the other hand, elevation and Tmax displayed almost flat trends across fault categories, further confirming their limited

relevance in predicting faults. These visual confirmations help strengthen the reliability of the statistical associations identified earlier.

4.4 Terrain Mapping

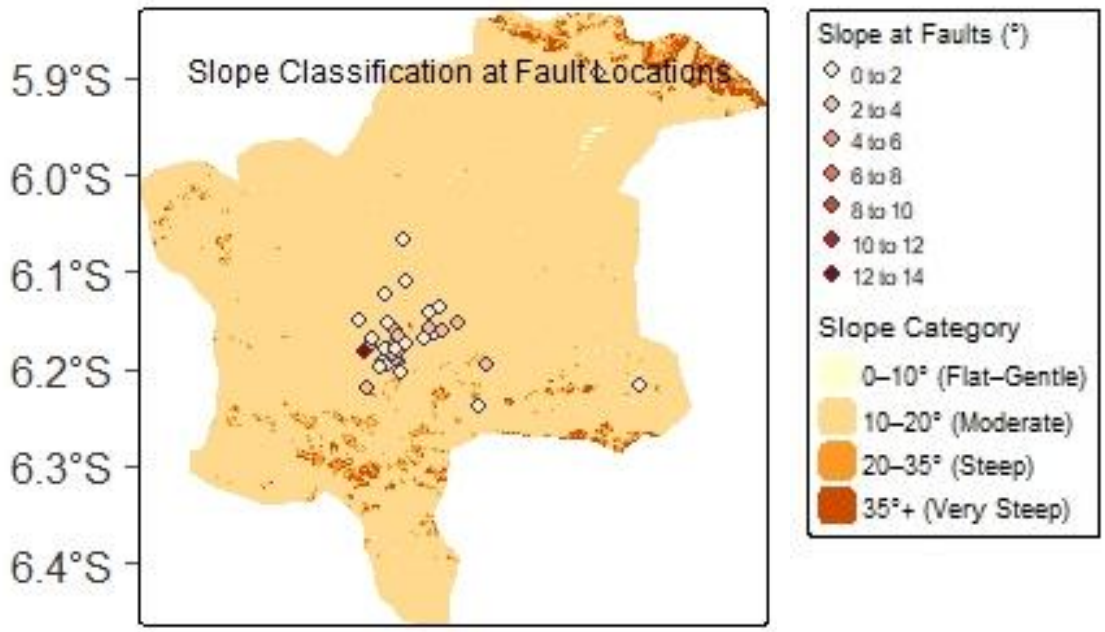

Figure 4.3: Terrain Mapping with Slope Classification at Fault Locations

This visualized map, as presented in Figure 4.3, provides a detailed visual analysis of slope variations at fault locations, which is highly relevant to the research. By categorizing slopes into Flat-Gentle, Moderate, Steep, and Very Steep; the map helps illustrate how underlying terrain influence's fault occurrence. Some of the faults appear to be concentrated in areas with steeper slopes, as indicated by darker colours, suggesting that steep, unstable terrains may influence an increase in the physical stress on the underground cables or infrastructure of liking, thereby increasing the risk of faults. Although we may note that most of the faults do in deed occur in areas with flat-gentle to moderate terrain suggesting otherwise. Therefore, by integrating this slope classification data with the fault records and environmental variables, enhances the predictive model of which then allows for more targeted maintenance strategies and improved allocation of repair resources in high-risk, geologically challenging areas.

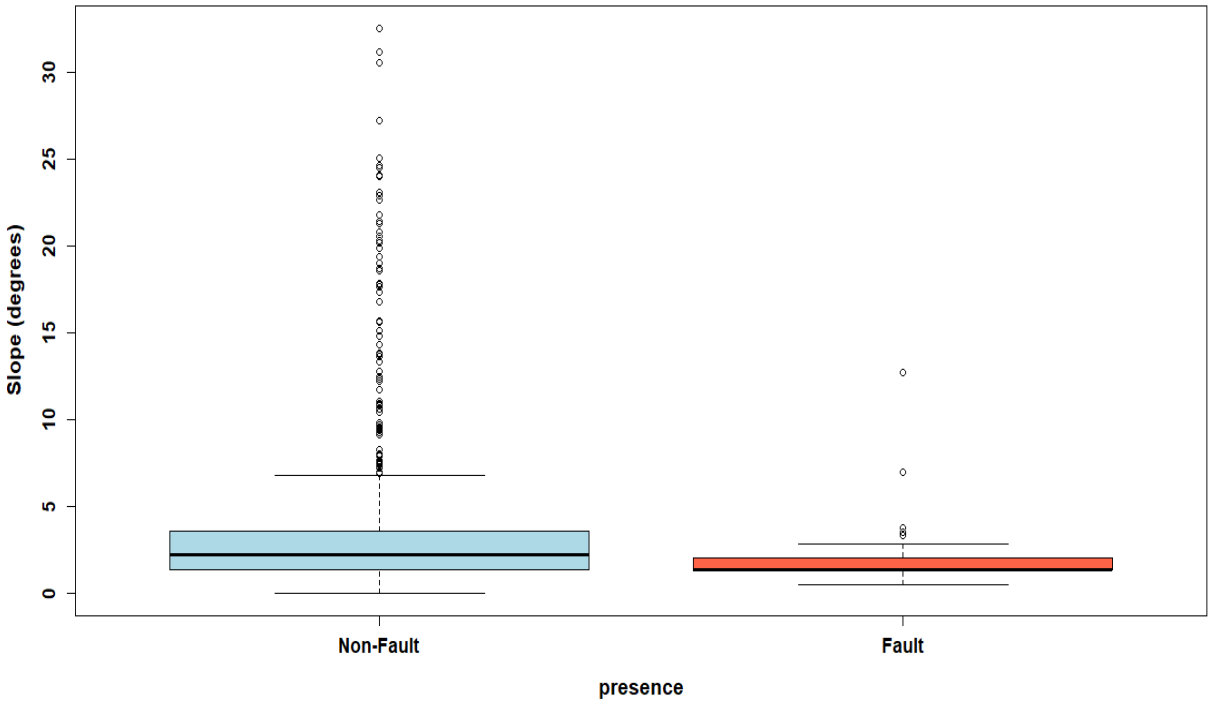

Figure 4.4: Box Plot of Slope Distribution in Fault Areas

The box plot presented in Figure 4.4, illustrates a clear difference between slope distributions in fault and non-fault areas, revealing critical terrain-related insights. In non-fault areas, the median slope is approximately 5°, with most values ranging between 0–10°, while occasional outliers extend beyond 10°, even reaching 30°, suggesting that steeper terrain generally experiences fewer faults. In contrast, fault areas show a markedly lower median slope near 0°, with values closely clustered between -2° to 2°, and very few outliers exceeding 8°, indicating that faults predominantly occur in flatter zones. This distribution highlights key fault-risk factors: flat terrain may suffer from soil instability, pooling effects, or inadequate drainage, increasing infrastructure vulnerability. Since faults are rare beyond 10°, maintenance efforts should focus on flat terrain where water-related stress could deteriorate underground ducts, ways to which underground fiber-optic cable are put in place, whereas aerial fiber-optic cables may be more susceptible to mechanical deformation in steeper regions. Understanding this dynamic allows TTCL to refine predictive maintenance strategies, directing resources to fault-prone areas based on terrain susceptibility.

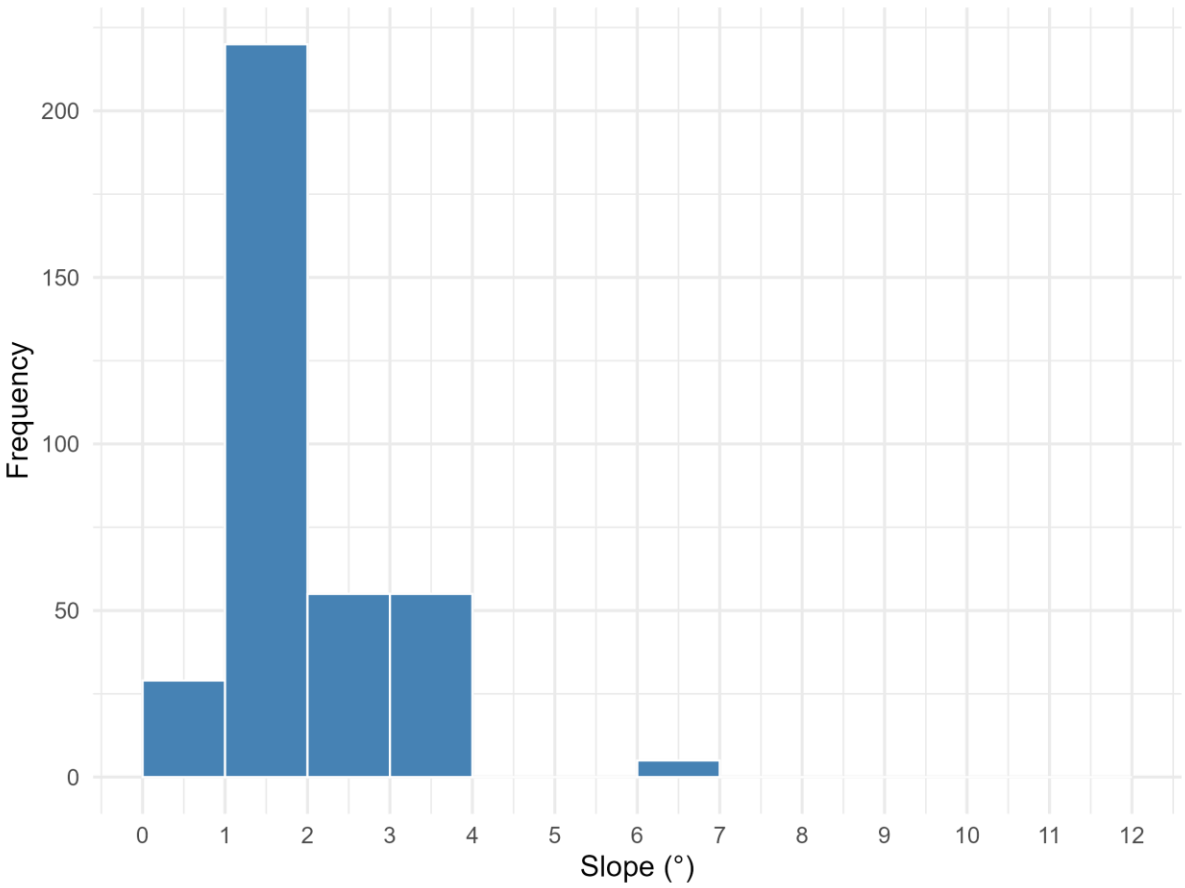

Figure 4.5: A Histogram of Fault Occurrence by Terrain Slope

This histogram, as pointed out by Figure 4.5, details the important relationship between fault locations and terrain slope, reinforcing previous findings on infrastructure susceptibility in low-gradient areas. This insight reveals that most faults occur in near-flat terrain (0–2° slope), with frequency exceeding 200 faults in this range. Beyond 4° slope, fault occurrence declines significantly, indicating that steeper terrain may apply less direct mechanical stress on fiber-optic infrastructure. This aligns with earlier fault vs. non-fault slope comparisons, confirming higher fault susceptibility in low-slope regions where soil instability and water pooling are potential risk factors. The distribution suggests that terrain-related influences on faults might be more hydrological than mechanical, with increase in moisture exposure weakening underground duct resilience. Considering the noticeable tendency of faults to occur in flat terrain, proactive maintenance should prioritize these areas, particularly locations prone to waterlogging or weak soil foundations. Incorporating terrain risk into TTCL's predictive framework would enhance pre-emptive inspection strategies and infrastructure resilience.

4.5 Fault Risk Mapping

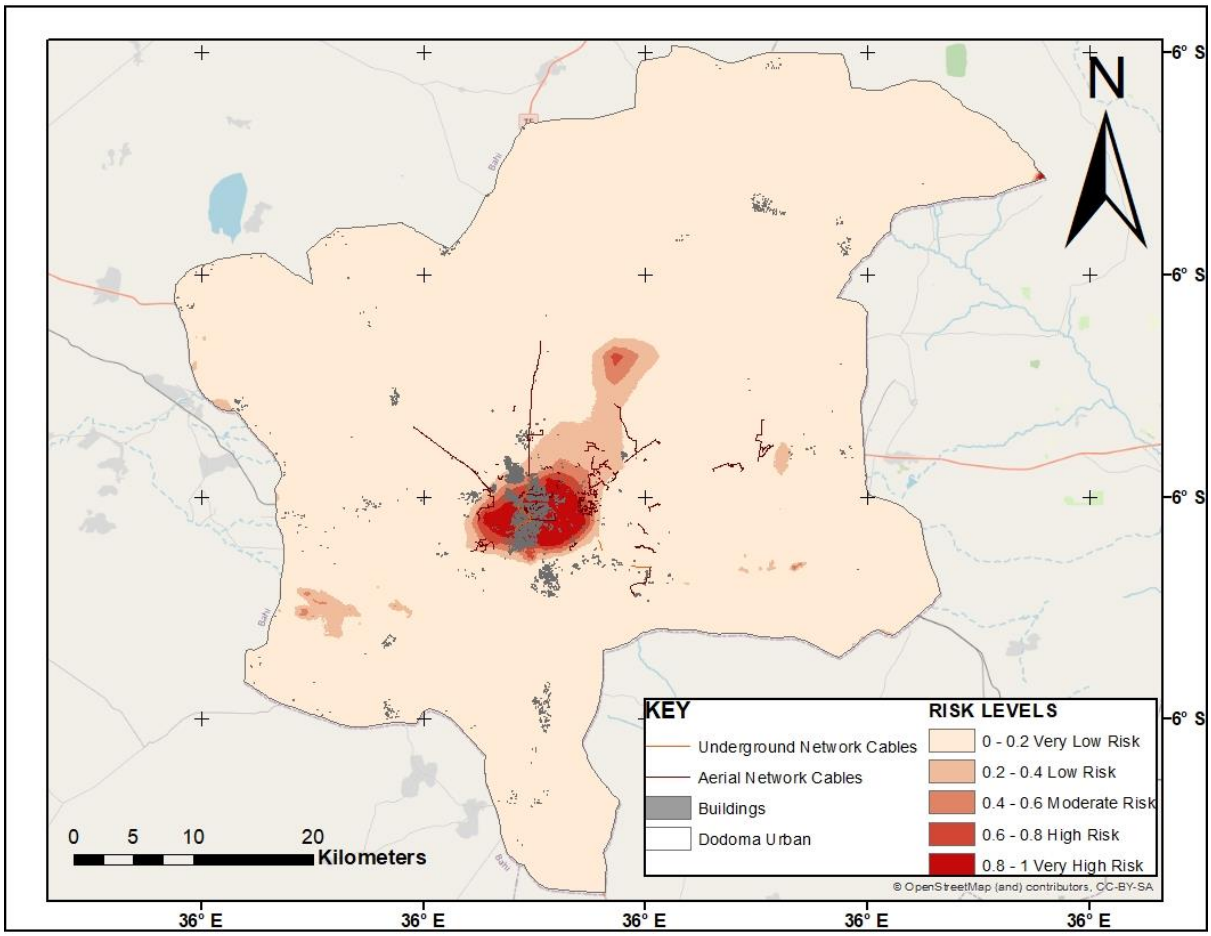

Figure 4.6: Fault Risk Map

The output, as shown by Figure 4.6, is demonstrating that the implemented predictive model is effective in identifying high-risk zones for network faults in Dodoma Urban. By mapping the risk levels through the use of a graduated colour spectrum; from low-risk areas in light shades to high-risk zones in darker tones, the map confirms the hypothesis that spatial and environmental data can be systematically integrated to predict fault-prone areas. This detailed spatial analysis clearly underlines that the variables such as terrain, weather patterns, and potentially other environmental factors significantly influence fault occurrences, reinforcing the shift from reactive to a proactive maintenance strategy. Basically, the map not only points out the critical regions or areas where network infrastructure is vulnerable but also tends to translates the complex data utilized into actionable insights for decision-makers in TTCL. It therefore supports the research's objective by highlighting operational priority areas, enabling TTCL to allocate resources more efficiently, optimize repair schedules, and reduce downtime.

4.6 Buffer Analysis: Evaluating Proximity Zones of Interest.

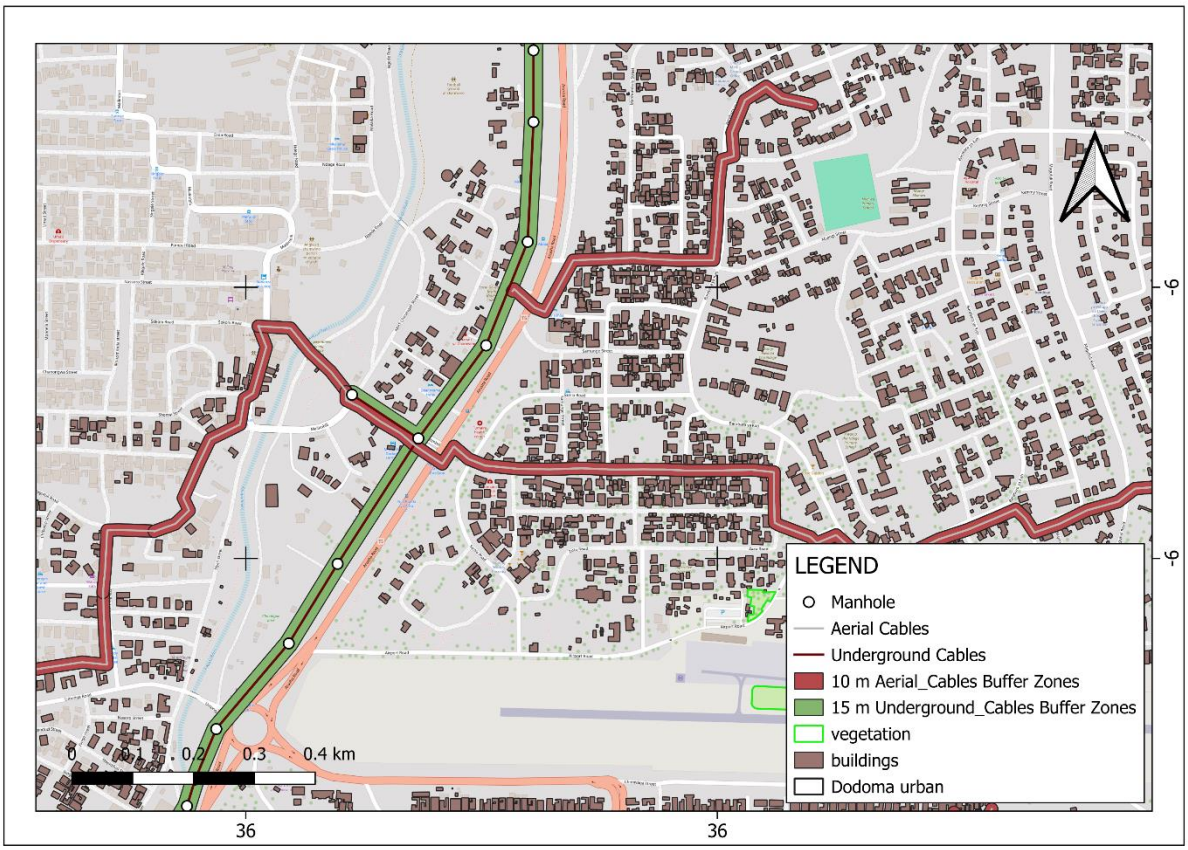

Figure 4.7: Buffer Zones Around Aerial & Underground Fiber-Optic Cables

Observation from Figure 4.7, shows the result of how proximity-based spatial analysis enhances fault risk prediction. With respect to the GIS field, the analysis helps in identifying how close faults risk can occur relatively to fixed assets like the observed underground or aerial fiber-optic cables. The logic behind this analysis is to determine whether there is a risk of exposing the aerial and underground fiber-optic cables to be affected and thus influencing the occurrence of fault existence and if influences the predicted high-risk fault locations to have fallen within the practical maintenance range from the existing network infrastructure. This insight informs TTCL's ability to prioritize inspections, improve route planning for repairs, and optimize resources by focusing on infrastructure segments closest to the identified risk zones. In this case, the buffer analysis tends to confirm that environmental and terrain risks do not act independently but are strongly influenced by spatial positioning relative to infrastructure. This insight of spatial proximity tends to strengthen the argument of adopting GIS in preventive infrastructure maintenance.

4.7 Predicted Faults Locations

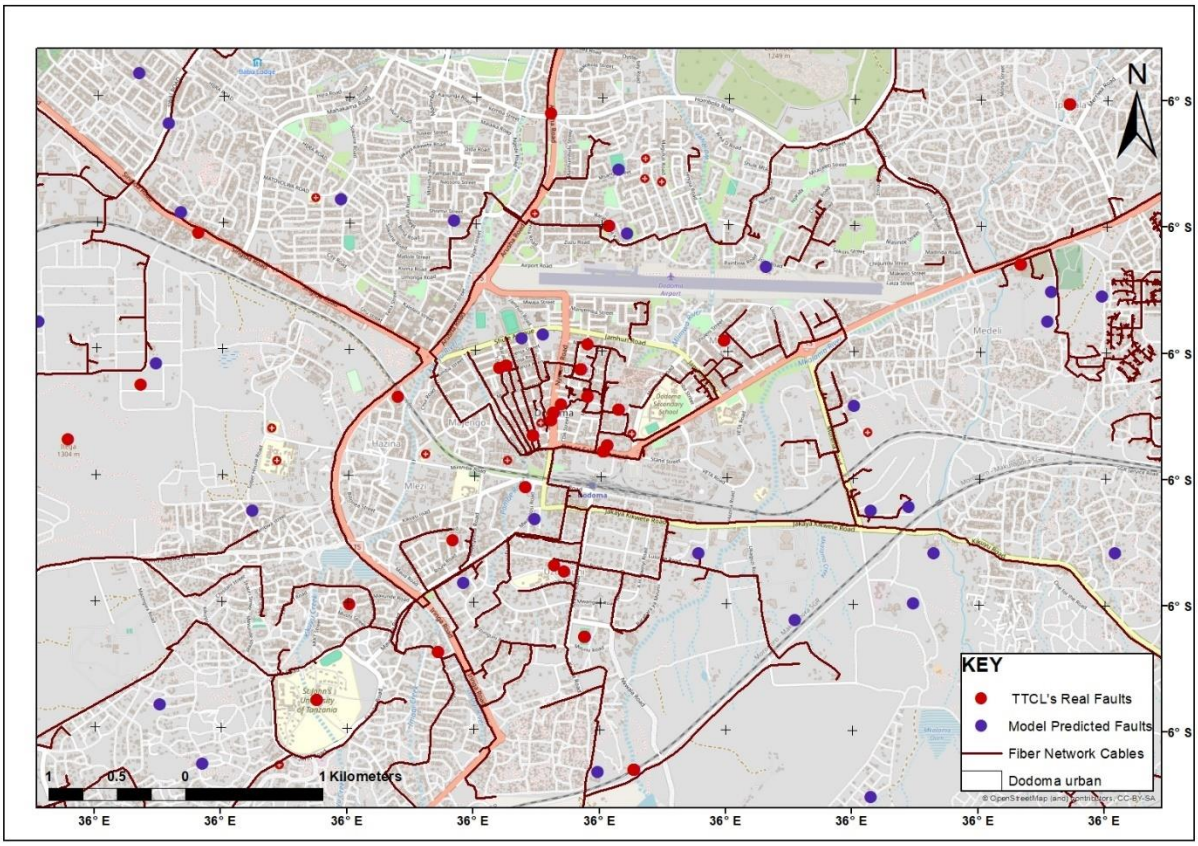

Figure 4.8: An Overlay of Real & Predicted Fault Locations

The introduced result, introduced by Figure 4.8, shows the spatial comparison of the locations of actual historical faults reported by TTCL with those predicted by the implemented logistic regression model, revealing a significant overlap in high-risk zones. This result confirms that the predictive model is not only statistically strong but also practically relevant. There's a high degree of spatial agreement between predicted and actual fault locations confirming that the environmental and spatial predictors such as rainfall, terrain, and temperature gradients are accurately captured by the model. Also, the spatial overlap between predicted and observed faults demonstrates strong compliance in flat zones, significantly aligning with the results in section 4.4 in particular.

CHAPTER FIVE
DISCUSSION

5.1 Overview

This chapter provides a critical interpretation of the results presented earlier in chapter four, emphasizing how the identified spatial and environmental predictors contribute to telecommunication faults. It connects the findings to the existing literature and demonstrating how the model's predictions align or diverge from previously published studies. It also explores the implications of the findings for TTCL's operational strategies, highlighting how a geospatial approach can enhance proactive fault mitigation by addressing both the strengths and limitations of the predictive model implemented and emphasizes how environmental, spatial, and infrastructure variables interact in predicting fault-prone zones. The discussion therefore provides us with evidence-based arguments and drawing important insights.

5.2 Discussion on Terrain Mapping

Terrain slope was one of the environmental variables analysed to assess its influence on fault distribution across the TTCL network. The slope layer was generated from a DEM, and its output in chapter four (Figure 4.3) showed that flatter terrain areas revealed a higher density of historical and predicted faults. This finding supports the hypothesis that areas with minimal slope are more prone to environmental accumulation, such as water pooling, which compromises underground fiber. This can be viewed in relation to Khosravi et al. (2020), where the infrastructure deployed in flat terrains faces increased risks due to topographic exposure and human interference. Cai (2002) emphasized the relevance of integrating slope data into spatial models for telecom planning, especially in urban areas where terrain subtly influences fault vulnerability.

5.3 Discussion on Predictive Model Performance and Accuracy

The Generalized Linear Model used for fault prediction as a predictive modelling approach in this research performed well as it achieved an AUC of 0.979 as shown in the ROC curve (Figure 3.10). This indicates a high level of sensitivity and specificity, meaning that the model was effective in distinguishing between fault-prone and non-fault-prone zones. Fawcett (2006) stated that an AUC value above 0.9 represents excellent model performance. The model's transparency allows decision-makers at TTCL to trace which variables influence fault probability, enhancing accountability and reliability. McCullagh and Nelder (1989) advocated the use of GLM in such real-world infrastructure modelling due to its interpretability and adaptability.

5.4 Discussion on Correlation Analysis

Correlation analysis was carried out to examine multicollinearity among variables, as detailed in chapter four (Figure 4.1). This analysis was crucial to ensure that strongly correlated predictors did not falsify the model performance. The correlation revealed that rainfall had high correlation and soil moisture had moderate correlation, which aligns with expectations since high precipitation increases ground saturation. Although soil moisture did not emerge as a statistically significant predictor in the final model, through its summary that was tabulated on Table 4.1, its correlation with rainfall suggests potential indirect effects. Naghibi and Pourghasemi (2015) observed that prolonged soil moisture may lead to gradual degradation of buried infrastructure due to insulation failure and increased vulnerability. We can also add that, the observed linear nature of correlation may have failed to capture the threshold or non-linear effects, indicating that future model implementations should explore alternative methods like partial correlations or non-linear algorithms. Otherwise, the analysis not only informed variable selection for the predictive model but also emphasized the importance of data quality and dimensional independence in improving model accuracy and interpretability.

5.5 Discussion on the Fault Risk Map Visualization

The raster-based fault risk map (Figure 4.6) visualized the spatial probability of fault occurrence across Dodoma Municipal. This output served as the core product of the predictive model implemented, revealing high-risk areas in the central and eastern zones of the study area. The map integrated environmental layers such as elevation, rainfall, and temperature to compute fault likelihood. Similar to findings by Ghorbanzadeh et al. (2019), spatial risk maps can enhance infrastructure resilience by identifying zones that require immediate intervention. These zones can be prioritized for surveillance, cable reinforcement, or relocation. Also, we may note that, even though the fault risk map effectively translates complex environmental data into actionable insights for maintenance planning, the map's scope is limited by its exclusion of operational variables like infrastructure age, human interference, or material type. Some predicted high-risk zones may not experience faults due to upgraded systems, while low-risk areas could still face faults driven by human or technical factors. Thus, the map produced and visualized is best used for regional planning, with its value maximized when integrated with TTCL's asset management systems and supplemented by routine updates, especially during critical seasons.

5.6 Discussion on the Buffer Zones Around the Fiber-Optic Networks

Buffer zones, as shown in chapter four (Figure 4.7), created around TTCL's fiber infrastructure were utilized to evaluate the proximity of the fault locations including those provided by TTCL and those predicted by the model as a probabilistic chance of occurrence, if we are to integrate them on the map generated. It is therefore that this analysis was essential to determine how closely the model outputs align with practical maintenance zones as well as gaining insights on possible infrastructural influences to fault occurence. Akdemir et al. (2016) used a similar approach to optimize fiber inspection intervals and recommended maintenance resource allocation based on buffer proximity of such kinds. This confirms the significant value of the spatial proximity evaluation in preventive infrastructure planning and maintenance strategies to better provision of services.

5.7 Discussion on the Predicted Fault Locations

The results obtained from the predictive model implemented showed a strong ability to identify areas within Dodoma Municipal that are likely to experience fiber-optic faults. The predicted fault locations closely matched the real, historical fault records provided by TTCL, which confirms that the model successfully captured the actual risk patterns present in the area. As shown in Figure 4.8, there is a visible spatial overlap between the predicted high-risk zones and the areas where faults had been previously recorded. This consistency suggests that environmental and spatial factors used in the model, especially rainfall, temperature extremes, and terrain elevation, are indeed good indicators of where faults are likely to occur. This strengthens the model's reliability in supporting proactive maintenance.

However, a few predicted fault locations appeared in areas where no actual faults had been reported before. These could be seen as the false positives, possibly pointing to locations where risk is high but faults haven't yet occurred, or they may have been unreported due to data gaps. On the other hand, some actual fault locations were not predicted by the model (false negatives), which may be due to non-environmental factors like vandalism, construction activities, or aging infrastructure that were not included in the model.

As Zhang et al. (2023) argue, the alignment between predicted and observed faults is crucial for justifying GIS-based infrastructure planning. Generally, with these insights, TTCL can refine its maintenance framework by prioritizing inspections in the emerging risk zones, and enhance its fault monitoring system giving a room of strengthening its fault detection method and shift from reactive repairs to proactive inspections.

CHAPTER SIX
CONCLUSION AND RECOMMENDATION

6.1 Overview

This chapter summarizes the main findings of the study in relation to the research objectives. It highlights the successful implementation of a predictive model that identified fault-prone areas in TTCL's fiber-optic network using environmental and spatial data. Key risk factors include rainfall, temperature extremes, and elevation. The model supports a shift from reactive to proactive maintenance. Based on these results, the chapter offers practical recommendations for improving network management and guiding future research and planning using GIS-based solutions.

6.2 Conclusion

This research has successfully shown how GIS and predictive modelling can be used to improve how TTCL manages fiber-optic network faults in Dodoma. By using GLM, the research was able to estimate the likelihood of faults based on environmental factors like rainfall, temperature, elevation, and soil moisture. The model proved to be highly effective, with an AUC score of 0.979, confirming its strong accuracy in identifying areas most at risk. Among all the factors, rainfall and temperature extremes had the strongest influence on fault occurrence, showing that environmental stress plays a big role in damaging network infrastructure. The results also showed that most faults tend to happen in flat areas, which may be more prone to waterlogging or ground instability. Overall, the study has met its main objectives by identifying the key causes of faults, predicting where they are most likely to occur, and providing a proactive way that can help TTCL act earlier and more effectively. This approach can help reduce service delays, improve maintenance planning, and support a shift from reactive to proactive infrastructure management.

6.3 Recommendation

To support proactive maintenance and improve service reliability with respect to the findings from this research, TTCL should implement the GIS-based predictive model applied in this study. Emphasis should be placed on flat to moderate terrain areas and areas experiencing high rainfall, which were identified to be most fault-prone. The model's effectiveness can be enhanced by incorporating real-time environmental data and additional risk factors such as infrastructure age or human activity. Finally, future studies should consider using machine learning algorithms, applying the model to other regions, and include socio-economic factors like urban density or vandalism to build a more comprehensive predictive framework.

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APPENDICES

APPENDIX 1: Predictive Modelling and Validation in R studio

APPENDIX 2: Data Cleaning, Geocoding, Georeferencing and Integration in R studio

APPENDIX 3: Terrain Analysis and Mapping in R studio

APPENDIX 4: Correlation Analysis in R studio

APPENDIX 1

This Appendix provides the full code in form of a script that was used in the predictive modelling and model validation as well visualization of the respective risk maps and graphs of statistical relevancy

```r
## ---------------------------------------------
## PREDICTIVE MODEL SCRIPT
## ---------------------------------------------

# === Load Required Libraries ===
library(sf)
library(terra)
library(dplyr)
library(tmap)
library(pROC)
library(ggplot2)
library(tidyr)

# === Load Spatial Data ===
faults <- st_read("D:/DATA & OUTPUTS/TTCL PROJECT/SHAPEFILE/Faults_DodomaUrban.shp")
dodoma_urban <- st_read("D:/DATA & OUTPUTS/TTCL PROJECT/SHAPEFILE/Dodoma_urban_reprojected.shp")
buildings <- st_read("D:/DATA & OUTPUTS/TTCL PROJECT/INFRASTRUCTURAL DATA/filtered_osm_dodoma/building
fiber <- st_read("D:/DATA & OUTPUTS/TTCL PROJECT/INFRASTRUCTURAL DATA/Geojson Files/fibers.geojson")

# === Load Environmental Rasters ===
elevation <- rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/ELEVATION/Dodoma_Elevation_Data.t
rainfall <- rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/WEATHER PATTERNS/NEW DATA/Rainfall
soil_moisture <- rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/WEATHER PATTERNS/NEW DATA/Soi
tmax <- rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/WEATHER PATTERNS/NEW DATA/Maximum Temp
```

```r
tmin <- rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/WEATHER PATTERNS/NEW DATA/Minimum Temp

# === Reproject All Layers to Common CRS ===
target_crs <- st_crs(dodoma_urban)
faults <- st_transform(faults, target_crs)
buildings <- st_transform(buildings, target_crs)
fiber <- st_transform(fiber, target_crs)

# Reproject rasters
elevation <- project(elevation, target_crs$wkt)
rainfall <- project(mean(rainfall), target_crs$wkt)
soil_moisture <- project(mean(soil_moisture), target_crs$wkt)
tmax <- project(mean(tmax), target_crs$wkt)
tmin <- project(mean(tmin), target_crs$wkt)

# === Resample All Layers to Common Grid ===
ref_raster <- elevation
rainfall <- resample(rainfall, ref_raster)
soil_moisture <- resample(soil_moisture, ref_raster)
tmax <- resample(tmax, ref_raster)
tmin <- resample(tmin, ref_raster)

# === Build Environmental Stack ===
env_stack <- c(elevation, rainfall, soil_moisture, tmax, tmin)
```

```r
env_stack <- c(elevation, rainfall, soil_moisture, tmax, tmin)
names(env_stack) <- c("elevation", "rainfall", "soil_moisture", "tmax", "tmin")
env_stack <- crop(env_stack, vect(dodoma_urban)) %>% mask(vect(dodoma_urban))

# === Sample Background Points ===
set.seed(123)
n_bg <- nrow(faults) * 3
bg_points <- spatSample(env_stack, n_bg, method = "random", as.points = TRUE, na.rm = TRUE)
bg_sf <- st_as_sf(bg_points)
bg_sf$presence <- 0

# === Tag Fault Points ===
faults <- faults[dodoma_urban, ]
faults$presence <- 1

# === Extract Raster Values to Training Data ===
faults_vect <- vect(faults)
bg_vect <- vect(bg_sf)

fault_vals <- terra::extract(env_stack, faults_vect) %>% select(-ID) %>% mutate(presence = 1)
bg_vals <- terra::extract(env_stack, bg_vect) %>% select(-ID) %>% mutate(presence = 0)

data_all <- bind_rows(fault_vals, bg_vals) %>% drop_na()

# === Train GLM Model ===
glm_model <- glm(presence ~ elevation + rainfall + soil_moisture + tmax + tmin,
                 data = data_all, family = binomial)
summary(glm_model)

# === Evaluate Model with ROC ===
data_all$predicted <- predict(glm_model, type = "response")
roc_obj <- roc(data_all$presence, data_all$predicted)
auc_val <- auc(roc_obj)
plot(roc_obj, col = "blue", main = "ROC Curve")

# Enhanced ROC Curve Visualization with AUC Annotation
# Create data frame from roc object
roc_df <- data.frame(
  fpr = rev(roc_obj$specificities),
  tpr = rev(roc_obj$sensitivities)
)

# Extract AUC
auc_val <- auc(roc_obj)

# Enhanced ROC plot with perfect classifier point
ggplot(roc_df, aes(x = 1 - fpr, y = tpr)) +
  geom_line(color = "blue", size = 1.2) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "brown") +
  geom_point(data = roc_df[which.max(roc_df$tpr + (1 - roc_df$fpr)), ],
             aes(x = 1 - fpr, y = tpr), color = "darkblue", size = 3) +
  # Add perfect classifier dot
  geom_point(aes(x = 0, y = 1), color = "blue", size = 4) +
  annotate("text", x = 0.02, y = 0.97, label = "PERFECT CLASSIFIER", hjust = 0, color = "blue", size =
  # AUC label and interpretive zones
  annotate("text", x = 0.6, y = 0.2, label = paste("AUC =", round(auc_val, 3)), size = 6, color = "dar
  annotate("text", x = 0.2, y = 0.85, label = "BETTER", size = 6, color = "black") +
  annotate("text", x = 0.8, y = 0.2, label = "WORSE", size = 6, color = "black") +
  labs(title = "ROC Curve",
       x = "False Positive Rate",
       y = "True Positive Rate") +
  theme_minimal(base_size = 14)

ggsave("D:/DATA & OUTPUTS/TTCL PROJECT/OUTPUTS/ROC_curve_AUC_with_perfect_classifier.png", width = 8,

# === Predict on Full Raster ===
risk_map <- predict(env_stack, glm_model, type = "response")
risk_map <- mask(risk_map, vect(dodoma_urban))

# === Export Risk Map ===
```

```r
# === Export Risk Map ===
writeRaster(risk_map, "D:/DATA & OUTPUTS/TTCL PROJECT/OUTPUTS/TTCL_fault_risk_map.tif", overwrite = TR

# ---------------------------------------------
# Extract Fault Points from Top 5% Hotspots ===
# ---------------------------------------------

# Step 1: Convert raster values to dataframe
risk_df <- as.data.frame(risk_map, xy = TRUE, na.rm = TRUE)
colnames(risk_df) <- c("x", "y", "risk")

# Step 2: Get cutoff for top 5%
top5_cutoff <- quantile(risk_df$risk, probs = 0.95, na.rm = TRUE)
top_risk_df <- risk_df %>% filter(risk >= top5_cutoff)

# Step 3: Randomly sample 100 points from top risk zones
set.seed(999)
sampled_points_df <- top_risk_df %>% sample_n(size = 100)

# Step 4: Convert to spatial features
predicted_faults_hotspots <- st_as_sf(sampled_points_df, coords = c("x", "y"), crs = st_crs(dodoma_urb

# Step 5: Visualize Results
```

```r
# Step 2: Get cutoff for top 5%
top5_cutoff <- quantile(risk_df$risk, probs = 0.95, na.rm = TRUE)
top_risk_df <- risk_df %>% filter(risk >= top5_cutoff)

# Step 3: Randomly sample 100 points from top risk zones
set.seed(999)
sampled_points_df <- top_risk_df %>% sample_n(size = 100)

# Step 4: Convert to spatial features
predicted_faults_hotspots <- st_as_sf(sampled_points_df, coords = c("x", "y"), crs = st_crs(dodoma_urb

# Step 5: Visualize Results
tm_shape(risk_map) +
  tm_raster(style = "cont", palette = "-RdYlBu", title = "Fault Risk") +
  tm_shape(predicted_faults_hotspots) +
  tm_dots(col = "black", size = 0.05, title = "Predicted Faults (Top 5%)") +
  tm_shape(dodoma_urban) + tm_borders(col = "black") +
  tm_layout(main.title = "TTCL Fault Hotspot Prediction", legend.outside = TRUE)

# Step 6: Save Output Shapefile
st_write(predicted_faults_hotspots,
         "D:/DATA & OUTPUTS/TTCL PROJECT/OUTPUTS/predicted_faults_top5_percent.shp",
         delete_dsn = TRUE)
```

APPENDIX 2

This Appendix provides a code that was used in the data pre-processing steps which include; Data cleaning, integration, georeferencing, and geocoding.

```r
# Load required libraries
library(readxl)
library(dplyr)
library(stringr)
library(tidygeocoder)
library(sf)

# Step 1: Load Excel file
file_path <- "D:/DATA & OUTPUTS/GIS DATA OUTPUTS/TTCL PROJECT/FAULT & CUSTOMER DATA/Fault report july-c
df <- read_excel(file_path)

# Step 2: Rename columns for convenience
df_clean <- df %>%
  rename(
    region = REGION_NAME,
    address = `Customer Address`
  )

# Step 3: Clean addresses and tag PO Boxes
df_clean <- df_clean %>%
  mutate(
    address = str_to_title(address),
    address = str_trim(address),
    is_po_box = str_detect(address, regex("P\\.?O\\.?|PO BOX|BOX", ignore_case = TRUE))
```

```r
# Step 3: Clean addresses and tag PO Boxes
df_clean <- df_clean %>%
  mutate(
    address = str_to_title(address),
    address = str_trim(address),
    is_po_box = str_detect(address, regex("P\\.?O\\.?|PO BOX|BOX", ignore_case = TRUE)),
    full_address = paste(address, region, "Tanzania", sep = ", ")
  )

# Step 4: Geocode
geocoded <- df_clean %>%
  geocode(address = full_address, method = 'osm', lat = latitude, long = longitude, limit = 1)

# Step 5: Convert to spatial object in WGS 84
geocoded_sf <- geocoded %>%
  filter(!is.na(latitude) & !is.na(longitude)) %>%
  st_as_sf(coords = c("longitude", "latitude"), crs = 4326)

# ✅ Step 6: Transform to UTM Zone 36S (EPSG:32736)
geocoded_sf_utm <- st_transform(geocoded_sf, crs = 32736)

# Step 7: Save results
write.csv(geocoded, "D:/DATA & OUTPUTS/GIS DATA OUTPUTS/TTCL PROJECT/FAULT & CUSTOMER DATA/fault_custom
```

```r
# Step 7: Save results
write.csv(geocoded, "D:/DATA & OUTPUTS/GIS DATA OUTPUTS/TTCL PROJECT/FAULT & CUSTOMER DATA/fault_custom

# Save as GeoPackage
st_write(
  geocoded_sf_utm,
  "D:/DATA & OUTPUTS/GIS DATA OUTPUTS/TTCL PROJECT/FAULT & CUSTOMER DATA/fault_customers_geocoded_july-
  delete_dsn = TRUE
)

# Optional: Plot
plot(geocoded_sf_utm["region"], main = "TTCL Fault Locations (EPSG:32736)")

# Ensure correct CRS and valid geometries
geocoded_sf_utm <- geocoded_sf %>%
  st_transform(32736) %>%
  st_make_valid()

# Extract coordinates
coords_df <- st_coordinates(geocoded_sf_utm) %>% as.data.frame()
colnames(coords_df) <- c("X", "Y")

```

```r
str(coords_df)
summary(coords_df)
head(coords_df)

library(ggplot2)
library(viridis)
library(dplyr)

# Remove duplicate coordinates
coords_df_unique <- coords_df %>% distinct()

# Confirm again
print(nrow(coords_df_unique))  # Should be less than 899

# Plot heatmap using only unique coordinates
ggplot(coords_df_unique, aes(x = X, y = Y)) +
  stat_density2d(
    aes(fill = after_stat(level), alpha = after_stat(level)),
    geom = "polygon", contour = TRUE
  ) +
  scale_fill_viridis(option = "inferno") +
  scale_alpha(range = c(0.3, 0.8), guide = FALSE) +
  labs(
    title = "TTCL Fault Density Heatmap",
```

19:1 (Top Level) R Script

```r
# Remove duplicate coordinates
coords_df_unique <- coords_df %>% distinct()

# Confirm again
print(nrow(coords_df_unique))  # Should be less than 899

# Plot heatmap using only unique coordinates
ggplot(coords_df_unique, aes(x = X, y = Y)) +
  stat_density2d(
    aes(fill = after_stat(level), alpha = after_stat(level)),
    geom = "polygon", contour = TRUE
  ) +
  scale_fill_viridis(option = "inferno") +
  scale_alpha(range = c(0.3, 0.8), guide = FALSE) +
  labs(
    title = "TTCL Fault Density Heatmap",
    x = "Easting (m)",
    y = "Northing (m)",
    fill = "Density"
  ) +
  theme_minimal() +
  coord_equal()
+ geom_point(data = coords_df_unique, aes(x = X, y = Y), color = "white", size = 0.5)

```

19:1 (Top Level) R Script

APPENDIX 3

This appendix provides the codes that were significant in the analysis of the terrain and visualization in terms of mapping

```r
# ========================
# TERRAIN ANALYSIS
# ========================
# Load Required Packages
library(terra)
library(sf)
library(tmap)
library(dplyr)
library(ggplot2)  # plotting histogram

# 1. Load data
elevation <- rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/ELEVATION/Dodoma_Elevation_Data.tif")
faults     <- st_read("D:/DATA & OUTPUTS/TTCL PROJECT/SHAPEFILE/Faults_DodomaUrban.shp")

# 2. Compute slope in degrees
slope <- terra::terrain(elevation, v = "slope", unit = "degrees", neighbors = 8)

# 3. Reclassify slope into 4 categories
m <- matrix(c(
  0,  10, 1,
  10, 20, 2,
  20, 35, 3,
  35, Inf,4
), ncol = 3, byrow = TRUE)
slope_class <- classify(slope, m)

```

```r

# 4. Label classes
slope_lvls <- c(
  "0–10° (Flat–Gentle)",
  "10–20° (Moderate)",
  "20–35° (Steep)",
  "35°+ (Very Steep)"
)

# 5. Extract slope at each fault point
# (terra::extract returns a data.frame with a column named "slope")
faults$slope <- terra::extract(slope, vect(faults))$slope

# 6. Build tmap
tmap_mode("plot")
slope_map <-
  tm_shape(slope_class) +
  tm_raster(
    palette    = "YlOrBr",
    labels     = slope_lvls,
    title      = "Slope Category"
  ) +
  tm_shape(faults) +
  tm_symbols(
    size       = 0.4,
    col        = "slope",
```

```r
    palette    = "Reds",
    title.col  = "Slope (°)"
  ) +
  tm_graticules(lines = FALSE, ticks = TRUE, col = "grey30") +
  tm_layout(
    title            = "Slope Classification at Fault Locations",
    title.position   = c("center","top"),
    legend.outside   = TRUE,
    legend.text.size = 1.2,
    legend.title.size= 1.4,
    frame            = TRUE
  )

# 7. Print & save the map
print(slope_map)
tmap_save(
  tm       = slope_map,
  filename = "D:/DATA & OUTPUTS/TTCL PROJECT/OUTPUTS/slope_fault_classified.png",
  width    = 10, height = 7, units = "in", dpi = 300
)

# BOXPLOT COMPARISON & HISTOGRAM OF SLOPE AT FAULT LOCATIONS #
# This Part Will Only Work After Running The Predictive Model Script First #
# --------- Extract Background Slopes (Optional for Comparison) ---------
bg_slope_vals <- terra::extract(slope, vect(bg_sf))
bg_slope_vals$presence <- 0
```

```r
fault_slope_vals <- terra::extract(slope, vect(faults))
fault_slope_vals$presence <- 1
slope_data <- bind_rows(fault_slope_vals, bg_slope_vals)

# -------- Histogram of slope at fault locations ---------
hist_plot <- ggplot(faults, aes(x = slope)) +
  geom_histogram(
    binwidth    = 1,
    boundary    = 0,
    closed      = "left",
    fill        = "steelblue",
    color       = "white"
  ) +
  scale_x_continuous(breaks = seq(0, 12, by = 1), limits = c(0, 12)) +
  labs(
    title = ,
    x     = "Slope (°)",
    y     = "Frequency"
  ) +
  theme_minimal(base_size = 14)

# --------- Print & save histogram -----------
print(hist_plot)
ggsave(
  filename = "D:/DATA & OUTPUTS/TTCL PROJECT/OUTPUTS/slope_fault_histogram.png",
  plot     = hist_plot,
```

```r
  plot     = hist_plot,
  width    = 8, height = 6, units = "in", dpi = 300
)


# --------- Boxplot Comparison with Enhanced Fonts ---------
par(
  font.main = 2,          # Bold title
  font.lab = 2,           # Bold axis labels
  font.axis = 2,          # Bold axis ticks
  cex.main = 1.4,         # Title size
  cex.lab = 1.3,          # Axis label size
  cex.axis = 1.2          # Axis tick size
)

boxplot(slope ~ presence, data = slope_data,
        names = c("Non-Fault", "Fault"),
        main = "Slope Comparison at Fault vs Non-Fault Areas",
        ylab = "Slope (degrees)",
        col = c("lightblue", "tomato"))

# Reset plotting parameters if needed (optional)
par(font.main = 1, font.lab = 1, font.axis = 1, cex.main = 1, cex.lab = 1, cex.axis = 1)

# --------- END OF TERRAIN ANALYSIS SCRIPT --------- #
```

APPENDIX 4

This appendix provides the codes necessary in the process of conducting correlation analysis vital for relationship visualization between variables of interest

```r
# ========================
# CORRELATION ANALYSIS
# ========================
# This Analysis Will Only Work After Running The Predictive Modelling Script

# Load Required Packages
library(dplyr)
library(corrplot)
library(ggplot2)

# 1. Select relevant columns from the dataset
cor_data <- data_all %>%
  select(presence, elevation, rainfall, soil_moisture, tmax, tmin)

# 2. Compute Pearson correlation matrix
cor_matrix <- cor(cor_data, use = "complete.obs", method = "pearson")

# 3. Visualize correlation matrix
corrplot(cor_matrix, method = "color",
         addCoef.col = "black",  # add numeric values to cells
         number.cex = 0.8,
         tl.cex = 0.9,
         title = "Pearson Correlation: Fault Presence vs Environmental Variables",
         mar = c(0, 0, 2, 0))

# 4. Extract and print correlations with fault presence
```

```r
# 4. Extract and print correlations with fault presence
cor_with_fault <- cor_matrix["presence", -1, drop = FALSE]
print(round(cor_with_fault, 3))

# 5. Add scatter plots for assessing relationship between env. variables and fault presence
# Load required libraries
library(ggplot2)
library(tidyr)

# Create a long-format version of the data for plotting
scatter_data <- data_all %>%
  select(presence, elevation, rainfall, soil_moisture, tmax, tmin) %>%
  pivot_longer(cols = -presence, names_to = "Variable", values_to = "Value")

# Plot scatterplots using ggplot2
ggplot(scatter_data, aes(x = Value, y = presence)) +
  # Use smaller, more transparent points with jitter
  geom_jitter(width = 0.2, height = 0.03, alpha = 0.3, color = "#1f77b4", size = 1.5) +

  # Add a linear trend line with confidence interval
  geom_smooth(method = "lm", color = "#d62728", se = TRUE, size = 1) +

  # Use facets for each variable, set free x-scale and more readable layout
  facet_wrap(~Variable, scales = "free_x", ncol = 3) +

  # Refine axis and titles
```

```r
  # Refine axis and titles
  labs(
    title = "Scatterplots: Environmental Variables vs Fault Presence",
    x = "Environmental Variable Value",
    y = "Fault Presence (0 = No, 1 = Yes)"
  ) +

  # Clean theme and tweak spacing
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    strip.text = element_text(face = "bold", size = 13),
    axis.title = element_text(face = "bold"),
    panel.spacing = unit(1.5, "lines")
  )

# PAIRWISE CORRELATION & DISTRIBUTION OF FAULTS VS VARIABLES #
# NB: All Data Where Reprojected To A Default CRS For Visualization In the Shiny App of R Unlike In the Pre
# ---------------- Load Required Libraries ----------------
library(sf)
library(terra)
library(dplyr)
library(pROC)
library(shiny)
library(shinythemes)
library(leaflet)
```

291:1 Visualize pairwise relationships with color indicating fault presence R Script

```r
library(leaflet)
library(RColorBrewer)
library(mapview)
library(webshot)
library(webshot2)
library(GGally)

# ---------------- Load Data ----------------
load_data <- function() {
  list(
    faults = st_read("D:/DATA & OUTPUTS/TTCL PROJECT/SHAPEFILE/Faults_DodomaUrban.shp") %>% st_transform(38
    dodoma_urban = st_read("D:/DATA & OUTPUTS/TTCL PROJECT/SHAPEFILE/Dodoma_urban_reprojected.shp") %>% st_
    elevation = rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/ELEVATION/Dodoma_Elevation_Data.tif
    rainfall = rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/WEATHER PATTERNS/NEW DATA/Rainfall_2
    soil_moisture = rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/WEATHER PATTERNS/NEW DATA/SoilM
    tmax = rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/WEATHER PATTERNS/NEW DATA/Maximum Temper
    tmin = rast("D:/DATA & OUTPUTS/TTCL PROJECT/ENVIRONMENTAL DATA/WEATHER PATTERNS/NEW DATA/Minimum Temper
    buildings = st_read("D:/DATA & OUTPUTS/TTCL PROJECT/INFRASTRUCTURAL DATA/filtered_osm_dodoma/buildings_
    fiber = st_read("D:/DATA & OUTPUTS/TTCL PROJECT/INFRASTRUCTURAL DATA/Geojson Files/fibers.geojson") %>%
  )
}

# ---------------- Build Model ----------------
build_model <- function(data) {
  data$faults$elevation <- terra::extract(data$elevation, vect(data$faults))[,2]
  data$faults$rainfall <- terra::extract(data$rainfall, vect(data$faults))[,2]
```

291:1 Visualize pairwise relationships with color indicating fault presence R Script

```r
  data$faults$soil_moisture <- terra::extract(data$soil_moisture, vect(data$faults))[,2]
  data$faults$tmax <- terra::extract(data$tmax, vect(data$faults))[,2]
  data$faults$tmin <- terra::extract(data$tmin, vect(data$faults))[,2]
  data$faults$building_density <- lengths(st_intersects(st_buffer(data$faults, 50), data$buildings))
  data$faults$fault_present <- 1

  set.seed(123)
  bg_points <- st_sample(data$dodoma_urban, 500, type = "random") %>% st_as_sf()
  bg_points <- st_transform(bg_points, 3857)
  bg_points$elevation <- terra::extract(data$elevation, vect(bg_points))[,2]
  bg_points$rainfall <- terra::extract(data$rainfall, vect(bg_points))[,2]
  bg_points$soil_moisture <- terra::extract(data$soil_moisture, vect(bg_points))[,2]
  bg_points$tmax <- terra::extract(data$tmax, vect(bg_points))[,2]
  bg_points$tmin <- terra::extract(data$tmin, vect(bg_points))[,2]
  bg_points$building_density <- lengths(st_intersects(st_buffer(bg_points, 50), data$buildings))
  bg_points$fault_present <- 0

  combined_data <- bind_rows(st_drop_geometry(data$faults), st_drop_geometry(bg_points))

  model <- glm(fault_present ~ elevation + rainfall + soil_moisture + tmax + tmin + building_density,
               data = combined_data, family = binomial())
  predicted_probs <- predict(model, type = "response")
  roc_obj <- roc(combined_data$fault_present, predicted_probs)

  list(model = model, roc = roc_obj, combined_data = combined_data)
}
```

291:1 Visualize pairwise relationships with color indicating fault presence R Script

```r
  bg_points$building_density <- lengths(st_intersects(st_buffer(bg_points, 50), data$buildings))
  bg_points$fault_present <- 0

  combined_data <- bind_rows(st_drop_geometry(data$faults), st_drop_geometry(bg_points))

  model <- glm(fault_present ~ elevation + rainfall + soil_moisture + tmax + tmin + building_density,
               data = combined_data, family = binomial())
  predicted_probs <- predict(model, type = "response")
  roc_obj <- roc(combined_data$fault_present, predicted_probs)

  list(model = model, roc = roc_obj, combined_data = combined_data)
}

# --------------- Load model data ---------------
model_data <- build_model(load_data())

# --------------- Select variables for a broader pairwise analysis ---------------
correlation_data <- model_data$combined_data %>%
  select(elevation, rainfall, soil_moisture, tmax, tmin, fault_present) %>%
  mutate(fault_present = as.factor(fault_present))  # ensure it's a factor

# --------------- Visualize pairwise relationships with color indicating fault presence ---------------
ggpairs(correlation_data, aes(color = fault_present),
        title = "Pairwise Correlation & Distribution: Faults vs Variables")

# END OF CORRELATION ANALYSIS SCRIPT #
```

293:1 Visualize pairwise relationships with color indicating fault presence R Script